

# DC Machines & Traction Motors

A complete learning and examination handbook covering DC-machine construction, generators, motors, starters, testing, speed control and electric-traction control—with derivations, diagrams, characteristic curves, solved numericals, Malayalam support, practice and full model answers.

COURSE CODE

**3032**

PROGRAMME

**Diploma in Electrical & Electronics Engineering**

SEMESTER

**3**

CREDITS

**3**

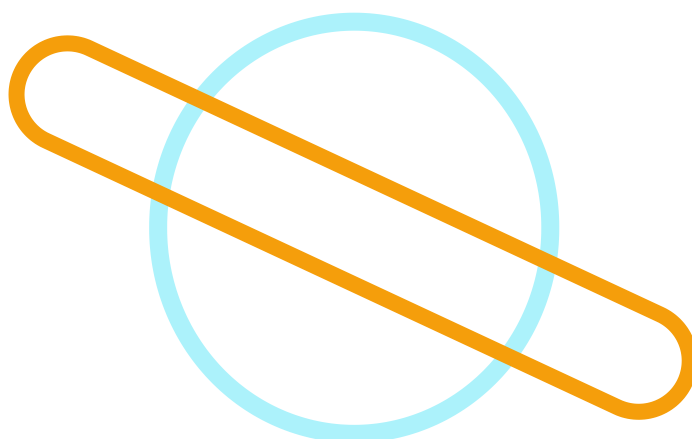
CATEGORY

**Program Core**

PERIODS

**3/week • 45/semester**[Start Learning](#)**PDF note**

The button opens the browser print dialog. Choose **Save as PDF** to create an offline A4 copy.



# Four-module pathway

## Module 1

Construction, generator action, EMF, losses, efficiency and windings

## Module 2

Armature reaction, commutation, characteristics and parallel operation

## Module 3

Motor action, torque, characteristics, starters and testing

## Module 4

Speed control, traction motors, transitions and electrical braking

# Why DC machines remain essential

DC machines convert energy in both directions. As generators they convert mechanical input into direct-current electrical output. As motors they convert DC electrical input into mechanical torque. Their transparent magnetic and electrical relationships make them ideal for learning electromechanical energy conversion, while traction applications demonstrate high starting torque, controlled acceleration and electrical braking.

## Energy conversion

Faraday's law explains generated EMF; force on a current-carrying conductor explains motor torque. The same physical machine can operate in either direction.

## Industrial relevance

DC motors are used where wide speed control, high starting torque or predictable torque-current relationships are required.

## Traction relevance

Series motors historically suit locomotives, trams and cranes because their torque is very high at low speed and heavy current.

## Typical applications

### Generators

Battery charging, laboratory supplies, exciters, electroplating and welding.

### Shunt motors

Lathes, fans, blowers, pumps and machine tools requiring nearly constant speed.

### Series motors

Traction, cranes, hoists and elevators requiring high starting torque.

### Compound motors

Presses, conveyors and rolling mills with changing load and better speed regulation.

# Learn, trace, calculate and practise

## 1. Understand

Read the physical principle before memorising any equation.

## 2. Trace

Follow magnetic flux, current, power and motion arrows in every diagram.

## 3. Calculate

Use the solved format: given data, formula, substitution, unit and interpretation.

## 4. Verify

Attempt module questions, model paper and answer-key checks without looking first.

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## OFFICIAL COURSE INFORMATION

# Objectives and prerequisite

## Official course objectives

1. To familiarize students with the construction and operation of DC machines.
2. To recognize the classification and applications of DC machines.
3. To analyze the performance of DC generators and DC motors.
4. To provide basic knowledge of electric traction systems.

## Prerequisite

**Topic:** Magnetism and Electromagnetism

**Course:** Elementary Concepts of Electrical Systems

**Semester:** 2

## How the prerequisite supports this course

### Magnetic circuit

Flux, reluctance and permeability explain the path of field flux through yoke, poles, air gap and armature.

### Electromagnetic induction

Faraday's law and Lenz's law explain generated EMF and back EMF.

### Force production

A current-carrying conductor in a magnetic field experiences force, producing motor torque.

### Fleming's rules

Right-hand rule gives generator direction; left-hand rule gives motor-force direction.

#### OFFICIAL OUTCOMES

## Course outcomes and CO-PO mapping

| CO  | Description                                                                       | Duration | Cognitive level |
|-----|-----------------------------------------------------------------------------------|----------|-----------------|
| CO1 | Identify the construction and operation of DC generator.                          | 10 hours | Applying        |
| CO2 | Identify the electrical characteristics and uses of DC generators.                | 10 hours | Applying        |
| CO3 | Select a DC motor for specific applications based on performance characteristics. | 12 hours | Applying        |
| CO4 | Choose various methods of speed control of DC motors and traction motors.         | 11 hours | Applying        |
| —   | Series Tests                                                                      | 2 hours  | —               |

### CO-PO mapping

| Course outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | Interpretation         |
|----------------|-----|-----|-----|-----|-----|-----|-----|------------------------|
| CO1            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO2            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO3            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |
| CO4            | 3   | —   | —   | —   | —   | —   | —   | Strongly mapped to PO1 |

#### Mapping scale

3 – Strongly mapped; 2 – Moderately mapped; 1 – Weakly mapped.

#### MODULE-WISE LEARNING ROADMAP

# Official outcomes, hours and cognitive levels

| Module outcome | Official description                                                                                 | Hours | Level         |
|----------------|------------------------------------------------------------------------------------------------------|-------|---------------|
| M1.01          | Illustrate the constructional details of DC machines.                                                | 3     | Understanding |
| M1.02          | Explain the principle of operation of a DC generator.                                                | 2     | Understanding |
| M1.03          | Derive EMF equation of DC generator and solve problems.                                              | 3     | Applying      |
| M1.04          | Compare the armature windings of a DC machine.                                                       | 2     | Understanding |
| M2.01          | Illustrate armature reaction and its effects in DC generators.                                       | 2     | Understanding |
| M2.02          | Explain commutation process in DC generator and methods for improving commutation.                   | 2     | Understanding |
| M2.03          | Illustrate the characteristics of various types of DC generators and solve problems.                 | 3     | Applying      |
| M2.04          | Outline parallel operation and uses of various DC generators.                                        | 3     | Understanding |
| Series Test I  |                                                                                                      | 1     | —             |
| M3.01          | Explain the working principle of DC motor and solve various motor parameters.                        | 4     | Applying      |
| M3.02          | Illustrate the performance characteristics of DC motors and select motors for specific applications. | 4     | Applying      |
| M3.03          | Illustrate the working of various types of starters.                                                 | 2     | Understanding |
| M3.04          | Outline the procedures for testing of DC motors.                                                     | 2     | Understanding |
| M4.01          | Identify different types of speed control of DC motors.                                              | 3     | Applying      |
| M4.02          | Summarize the use of DC series motors in electric traction.                                          | 2     | Understanding |
| M4.03          | Illustrate speed control of traction motors.                                                         | 3     | Understanding |
| M4.04          | Explain braking of traction motors.                                                                  | 3     | Understanding |
| Series Test II |                                                                                                      | 1     | —             |

# Construction and Operation of DC Generator

M1.01–M1.04: construction, generator principle, EMF equation, losses, efficiency and armature windings.

## Electrical-machine idea

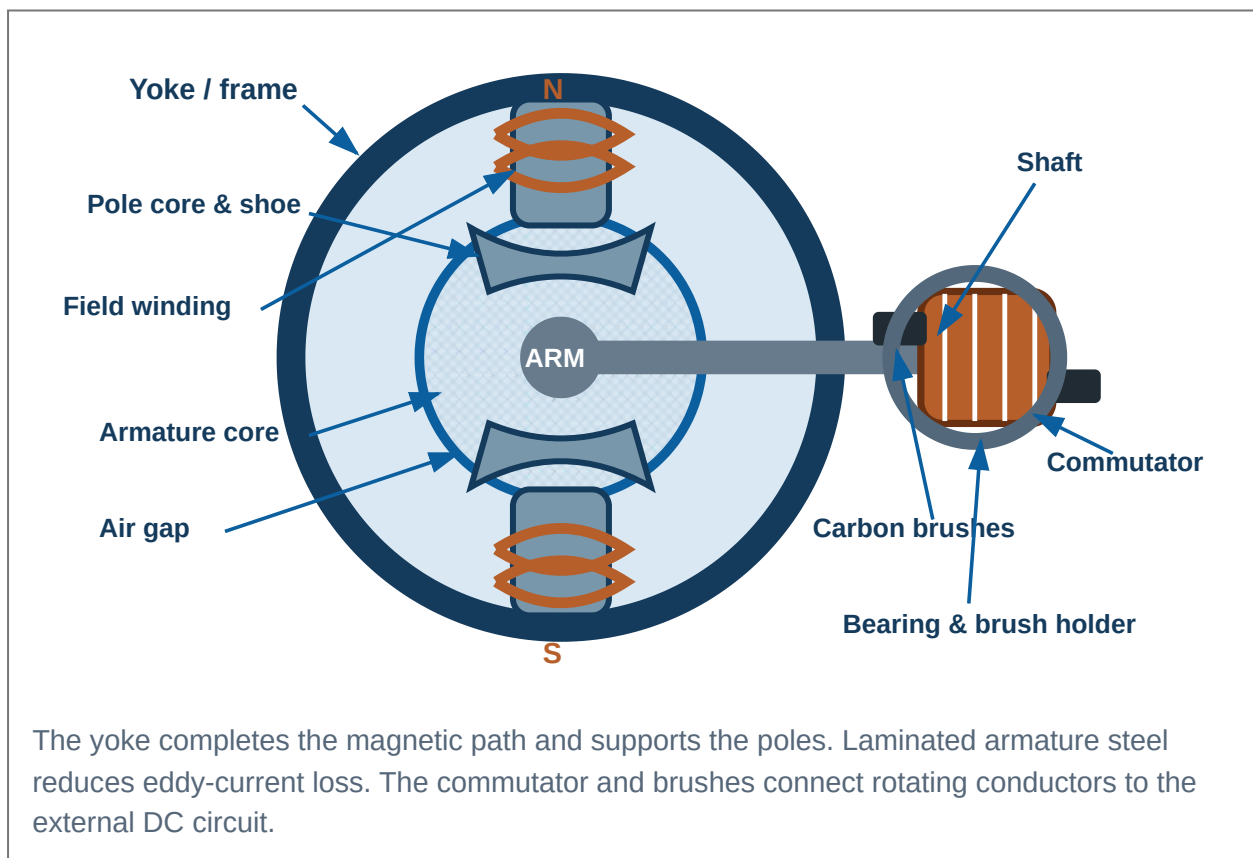
An electrical machine links an electrical system and a mechanical system through a magnetic field. In generator operation, a prime mover rotates conductors and electrical power is obtained. In motor operation, electrical current produces torque and mechanical output.

## Reversibility

The physical DC machine is reversible. The same armature, field system, commutator and brushes can operate as a generator or a motor; only the direction of energy flow changes.

## 1. Constructional details of a DC machine

The stationary field system creates the main magnetic flux. The rotating armature carries conductors in which EMF is induced or current flows. The commutator and brushes form the mechanical switching system that provides DC at the terminals and unidirectional torque.





| Part                 | Main purpose                                                  | Typical material / construction                                     |
|----------------------|---------------------------------------------------------------|---------------------------------------------------------------------|
| Yoke                 | Mechanical support and low-reluctance return path for flux    | Cast iron for small machines; rolled/cast steel for larger machines |
| Pole core            | Carries field winding and conducts flux                       | Laminated or solid steel depending on size                          |
| Pole shoe            | Spreads flux over armature, reduces reluctance, supports coil | Laminated steel                                                     |
| Field winding        | Produces main magnetomotive force                             | Insulated copper conductor                                          |
| Armature core        | Provides slots and magnetic path                              | Thin silicon-steel laminations keyed to shaft                       |
| Armature winding     | Carries generated/current-carrying conductors                 | Insulated copper coils placed in slots                              |
| Commutator           | Mechanical rectifier/current reverser                         | Hard-drawn copper segments insulated by mica                        |
| Brushes              | Sliding electrical contact with commutator                    | Carbon/graphite or copper-graphite                                  |
| Shaft & bearings     | Transmit torque and support rotation                          | Steel shaft with ball/roller bearings                               |
| Interpoles           | Improve commutation by neutralising reactance voltage         | Small poles between main poles, series with armature                |
| Compensating winding | Neutralises armature reaction under pole faces                | Conductors embedded in pole shoes, series with armature             |

## Why laminations are used

Alternating flux in armature teeth and core induces eddy currents. Dividing the core into insulated thin laminations greatly increases the resistance of unwanted current paths and reduces eddy-current heating. Silicon steel also reduces hysteresis loss.

## Cooling and mechanical support

Ventilation ducts, fans and open frames remove heat. End brackets, bearings, shaft and yoke preserve a uniform air gap. Poor alignment creates noise, vibration, brush sparking and unequal magnetic pull.

## 2. Magnetic circuit and generator principle

Main flux leaves the north pole, crosses the air gap, enters the armature, divides through the armature core, crosses the opposite air gap into the south pole and returns through the yoke. When armature conductors cut this flux, an EMF is induced according to Faraday's law.

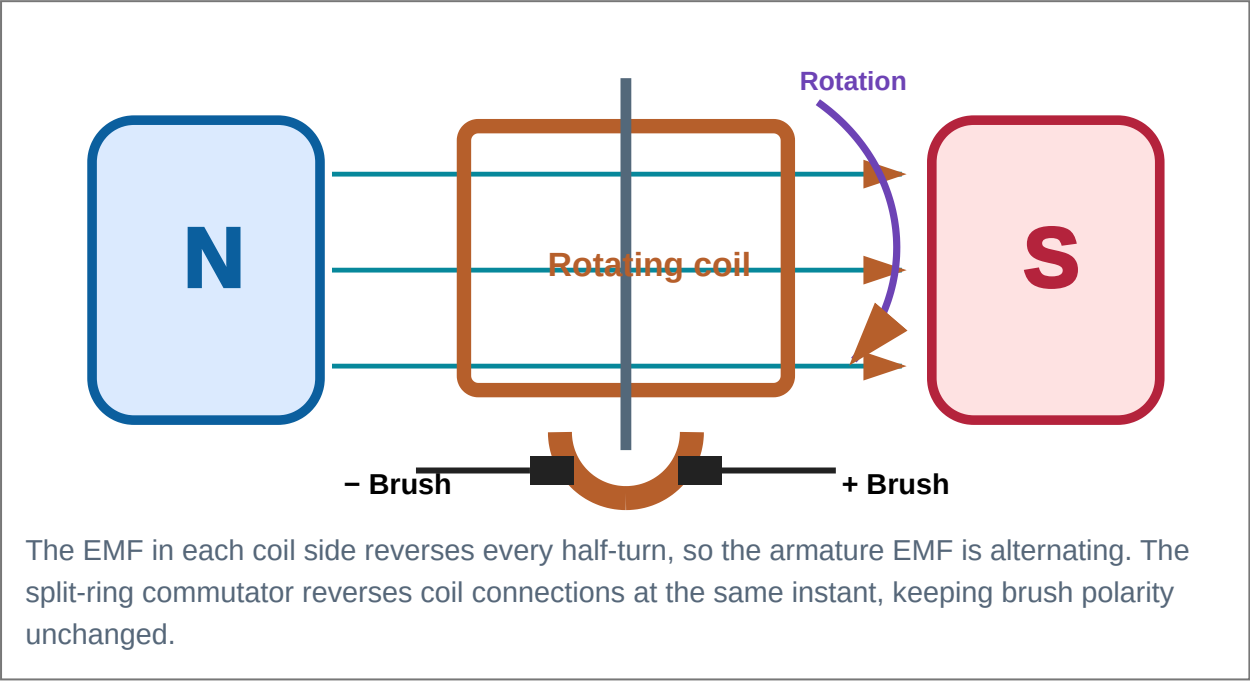
$$e = -N_c \frac{d\Phi}{dt}$$

The negative sign represents Lenz's law: induced EMF opposes the cause producing it.

Generator action

Mechanical rotation causes conductors to cut magnetic flux. Fleming's right-hand rule gives the induced-current direction: first finger = field, thumb = motion, middle finger = induced current.

Malayalam Support: ഏകദിശീയമായ ഏകാന്തതയിൽ ഏകദിശീയമായി  
ഏകദിശീയമായി ഏകദിശീയമായി EMF ഏകദിശീയമായി. ഏകദിശീയമായി  
induced current-ഏകദിശീയമായി ഏകദിശീയമായി.

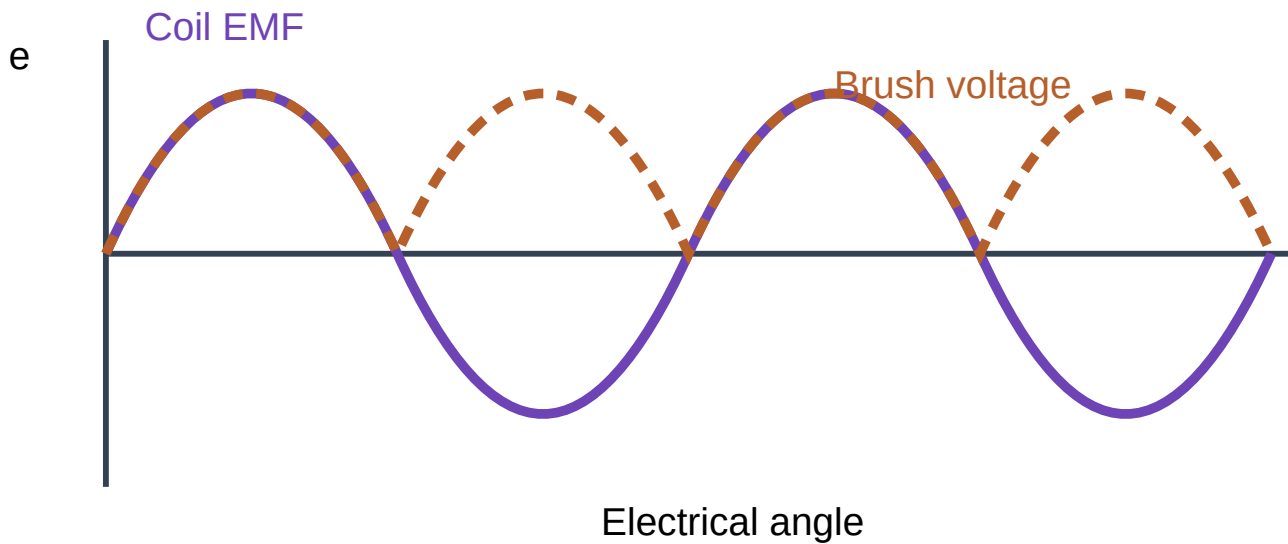


3. Simple-loop waveform and commutator rectification

Coil positions during one revolution

| Angle | Flux-cutting rate | Coil EMF | Brush output |
|-------|-------------------|----------|--------------|
| 0°    | Zero              | 0        | 0            |
| 90°   | Maximum           | +Emax    | +Emax        |
| 180°  | Zero              | 0        | 0            |
| 270°  | Maximum opposite  | -Emax    | +Emax        |
| 360°  | Zero              | 0        | 0            |

### Generated and rectified waveform



## 4. Classification by field excitation

### Separately excited

Field winding receives current from an independent DC supply. Flux can be controlled independently of armature load.

### Shunt

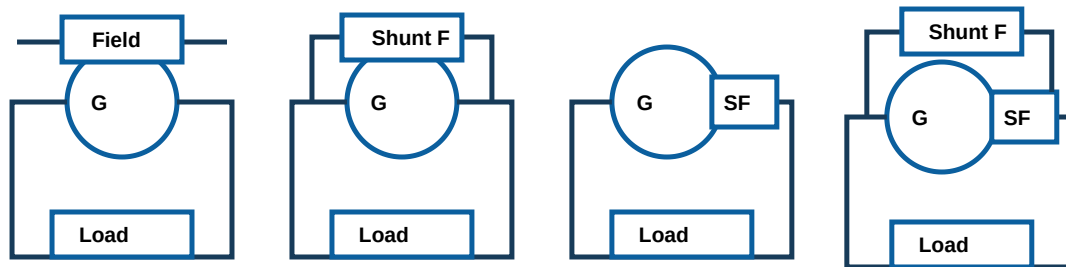
Field winding is connected across armature terminals. It has many turns, high resistance and small field current.

### Series

Field winding is in series with armature and load. It carries load current and uses few thick turns.

### Compound

Both shunt and series fields are used. Cumulative compounding aids flux; differential compounding opposes it.



Separately excited

Shunt

Series

Compound

SF denotes series field. Compound generators may be long-shunt or short-shunt depending on whether the shunt field is connected across both armature and series field or across the armature alone.

| Type                  | Field connection                 | Voltage behaviour                         | Typical use                                |
|-----------------------|----------------------------------|-------------------------------------------|--------------------------------------------|
| Separately excited    | Independent supply               | Wide controllability                      | Laboratory and controlled DC supply        |
| Shunt                 | Across terminals                 | Approximately constant voltage            | Battery charging, lighting, exciters       |
| Series                | Series with load                 | Voltage rises with load before saturation | Booster and special constant-current work  |
| Cumulative compound   | Series field aids shunt field    | Flat or rising external characteristic    | Distribution feeders, heavy load variation |
| Differential compound | Series field opposes shunt field | Voltage falls rapidly with load           | Rare; special drooping-voltage duty        |

## 5. Generator voltage equations

### Basic generator equation

$$V = E_g - I_a R_a - V_b$$

If brush drop is neglected,  $V = E_g - I_a R_a$ . State the assumption in numerical work.

### Current relations

**Shunt:**  $I_a = I_L + I_{sh}$ ,  $I_{sh} = V/R_{sh}$ .

**Series:**  $I_a = I_L$ ;  $V = E_g - I_a(R_a + R_{se})$ .

**Long-shunt compound:**  $I_{sh} = V/R_{sh}$ ;  $I_a = I_L + I_{sh}$ .

## 6. Derivation of the DC-generator EMF equation

- 1 Flux cut by one conductor in one complete revolution =  $P\Phi$  webers, because it passes under  $P$

poles.

2 Time for one revolution at N rpm =  $60/N$  seconds.

3 Average EMF induced in one conductor = flux cut / time =  $P\Phi N/60$  volts.

4 Conductors in series in each parallel path =  $Z/A$ .

5 Generated EMF between brushes = EMF per conductor  $\times$  conductors in series.

$$E_g = P\Phi ZN / 60A$$

| Symbol | Meaning                   | SI unit                |
|--------|---------------------------|------------------------|
| $E_g$  | Generated EMF             | volt                   |
| $P$    | Number of poles           | dimensionless          |
| $\Phi$ | Flux per pole             | weber                  |
| $Z$    | Total armature conductors | dimensionless          |
| $N$    | Speed                     | revolutions per minute |
| $A$    | Parallel paths            | dimensionless          |

Simplex lap winding:  $A = P$  | Simplex wave winding:  $A = 2$

### Solved numerical M1-1: Generated EMF

**Problem:** A 4-pole wave-wound generator has 600 conductors, flux 25 mWb/pole and speed 900 rpm. Find  $E_g$ .

**Given:**  $P=4$ ,  $Z=600$ ,  $\Phi=0.025$  Wb,  $N=900$  rpm,  $A=2$ .

**Formula:**  $E_g = P\Phi ZN / (60A)$ .

**Substitution:**  $E_g = (4 \times 0.025 \times 600 \times 900) / (60 \times 2) = 450$  V.

**Final answer:** Generated EMF = **450 V**. The wave winding gives only two parallel paths and therefore higher voltage.

**Common mistake:** Do not use  $A=P$  for a wave winding.

### Solved numerical M1-2: Speed from generated EMF

**Problem:** A 6-pole simplex lap generator has 720 conductors and 30 mWb/pole. Find speed for  $E_g=216$  V.

**Given:**  $P=6$ ,  $A=6$ ,  $Z=720$ ,  $\Phi=0.03$  Wb,  $E_g=216$  V.

**Formula:**  $N=60AE_g/(P\Phi Z)$ .

**Calculation:**  $N=(60 \times 6 \times 216)/(6 \times 0.03 \times 720)=600$  rpm.

**Final answer:** 600 rpm.

## 7. Losses, power stages and efficiency

### Copper losses

Armature:  $I_a^2 R_a$ . Shunt field:  $V^2/R_{sh}$  or  $V I_{sh}$ . Series field:  $I_a^2 R_{se}$ . Brush-contact loss may be taken as  $V_b I_a$  when specified.

### Iron losses

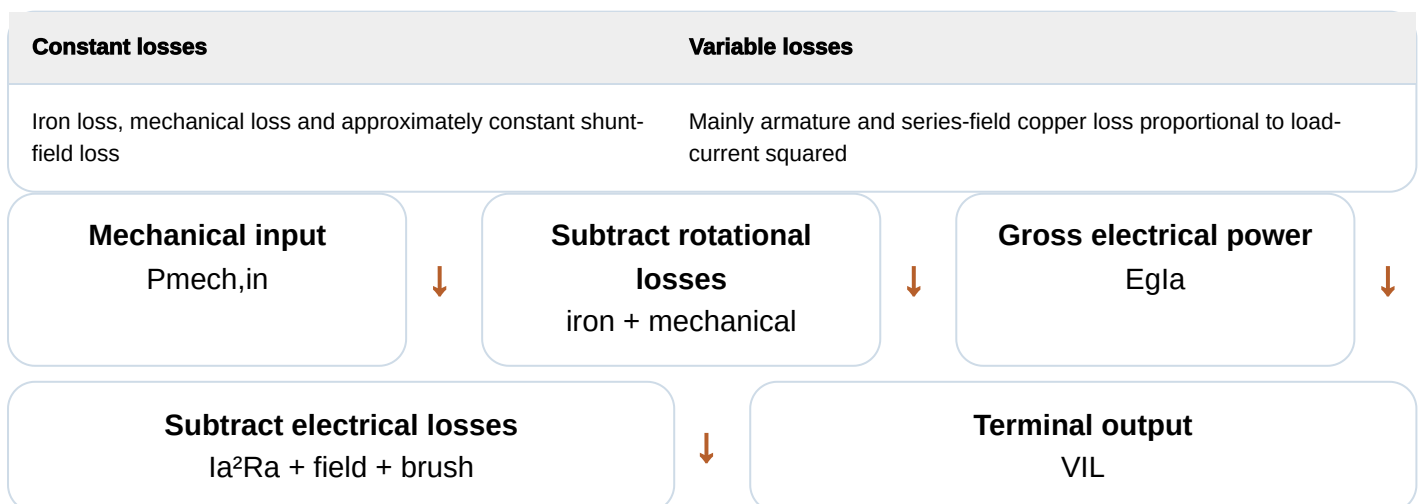
Hysteresis and eddy-current losses occur mainly in the armature core because its flux direction changes as it rotates.

### Mechanical losses

Bearing friction, brush friction and windage. They are approximately constant over the normal operating range.

### Stray-load loss

Additional loss caused by leakage flux, armature reaction and nonuniform current distribution. It is often estimated as a small percentage.



$$\eta_g = \text{Electrical output} / \text{Mechanical input} \times 100\%$$

## 8. Condition for maximum efficiency

- 1 Let terminal voltage be approximately constant, load current be  $I$  and effective variable resistance be  $R$ . Let constant loss be  $W_c$ .
- 2 Output =  $VI$ . Total loss =  $I^2R + W_c$ .
- 3 Efficiency  $\eta = VI / (VI + I^2R + W_c)$ .
- 4 For maximum efficiency, differentiate  $\eta$  with respect to  $I$  and set  $d\eta/dI=0$ .
- 5 After simplification:  $I^2R = W_c$ .

At maximum efficiency: variable copper loss = constant loss

This result assumes approximately constant terminal voltage, constant field loss and constant rotational losses. For a shunt generator, armature current rather than load current should be used when field current is appreciable.

### Solved numerical M1-3: Load at maximum efficiency

**Problem:** A 220 V shunt generator has armature resistance  $0.20 \Omega$  and constant losses 800 W. Neglect field-current difference. Find load current at maximum efficiency.

**Formula:**  $I^2R_a = W_c$ .

**Calculation:**  $I = \sqrt{(800/0.20)} = \sqrt{4000} = 63.25 \text{ A}$ .

**Output:**  $220 \times 63.25 = 13.915 \text{ kW}$ .

**Final answer:** Maximum efficiency occurs at approximately **63.3 A** load current.

## 9. Armature winding fundamentals

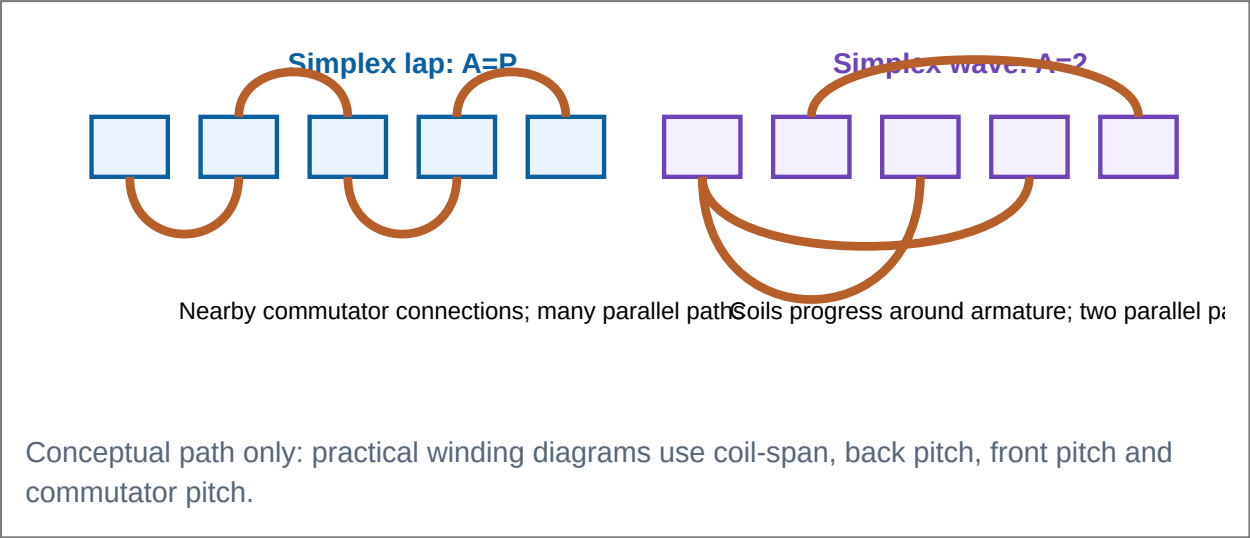
A coil has one or more turns. Each turn has two active coil sides placed in slots. Coil ends connect to commutator segments. The winding must provide closed paths and correctly spaced connections so generated EMFs add.

### Simplex lap winding

Each coil "laps back" to a nearby commutator segment. Parallel paths  $A=P$ . Brush sets are normally equal to pole number. Suitable for low-voltage, high-current machines.

Simplex wave winding

The winding progresses around the armature in a wave, connecting conductors under poles of similar polarity. Parallel paths  $A=2$ . Suitable for high-voltage, lower-current machines.



| Feature                     | Lap winding                             | Wave winding                                |
|-----------------------------|-----------------------------------------|---------------------------------------------|
| Parallel paths, simplex     | $A=P$                                   | $A=2$                                       |
| Brush sets                  | Usually $P$                             | Usually 2                                   |
| Voltage/current suitability | Low voltage, high current               | High voltage, low current                   |
| Equalizer rings             | Required to prevent circulating current | Normally not required                       |
| Dummy coil                  | Not normally required                   | May be required to satisfy pitch conditions |
| Typical use                 | Heavy-current generators                | Lighting and higher-voltage generators      |



## Module 1 expected-question practice

### One word

**Name the part acting as a mechanical rectifier.**

**Answer:** Commutator.

### Important diagram

**Draw and label DC-machine construction.**

Include yoke, pole core, pole shoe, field winding, armature, commutator, brushes, shaft and bearings.

### Important derivation

**Derive  $E_g = P\Phi ZN/60A$ .**

Use flux cut per revolution, time per revolution and conductors in series per path.

### Important numerical

**Find  $E_g$ ,  $N$ ,  $\Phi$  or  $Z$  from the EMF equation.**

First identify lap/wave winding and correct  $A$ .

### Comparison

**Compare lap and wave windings.**

Parallel paths, brushes, current/voltage rating and equalizer rings.

### Long answer

**Explain losses, power flow and maximum efficiency.**

Classify losses, draw stages and derive  $I^2R = W_c$ .

# Characteristics and Applications of DC Generators

M2.01–M2.04: armature reaction, commutation, generator characteristics, voltage build-up, parallel operation and applications.

## 1. Armature reaction

When load current flows in armature conductors, the armature produces its own magnetic field. This armature field combines with the main pole field and distorts the air-gap flux distribution. The effect of armature magnetomotive force on the main field is called **armature reaction**.

### Geometrical neutral axis (GNA)

The axis midway between adjacent pole centres. It is fixed by machine geometry.

### Magnetic neutral axis (MNA)

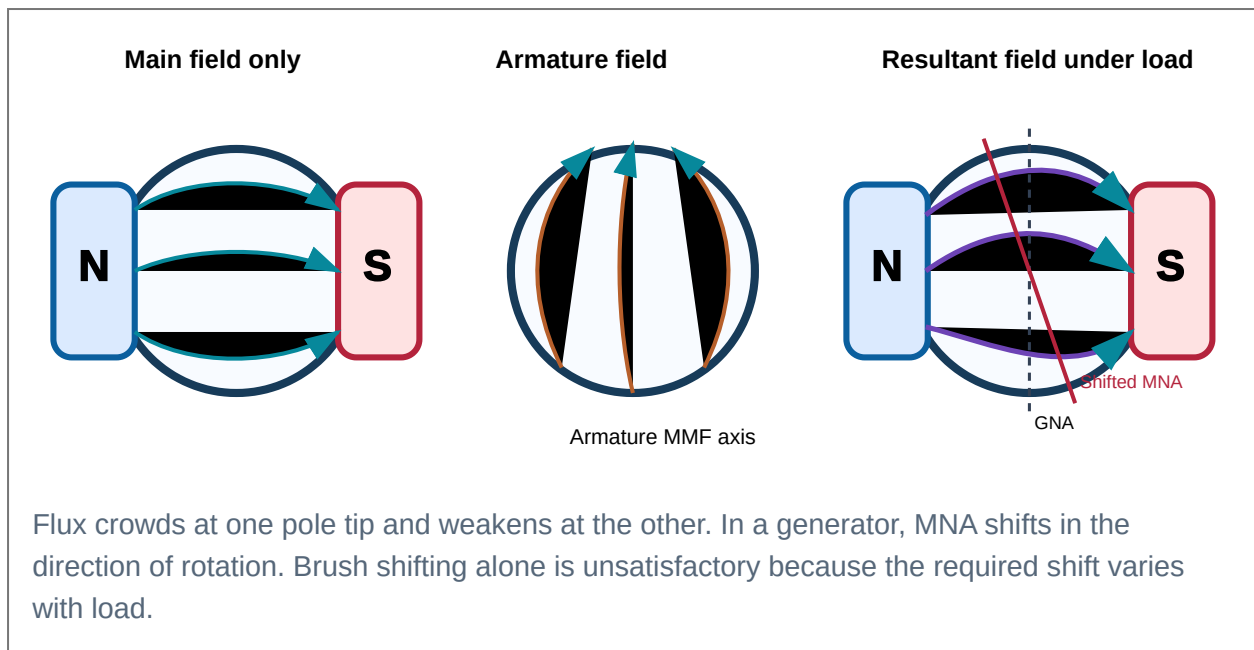
The axis along which armature conductors have zero induced EMF. Under load it shifts because the resultant field is distorted.

### Cross-magnetising effect

Armature MMF acts mainly across the pole axis, strengthening flux at one pole tip and weakening it at the other.

### Demagnetising effect

If brushes are shifted from the GNA, part of armature MMF directly opposes the main field and reduces average flux.



### Effects of armature reaction

Distorted flux, reduced average flux, lower generated EMF, shifted MNA, brush sparking, local heating at pole tips and poor commutation.

**Malayalam Support:** ഏതെങ്കിലും ഏതെങ്കിലും armature ഏതെങ്കിലും magnetic field

ഏതെങ്കിലും main field-ഏതെങ്കിലും neutral axis ഏതെങ്കിലും; ഏതെങ്കിലും brush sparking ഏതെങ്കിലും.

## 2. Interpoles and compensating windings

### Interpoles

Small poles placed midway between main poles. Their windings are connected in series with the armature, so interpole flux automatically varies with load. In a generator, interpole polarity is the same as the next main pole in the direction of rotation. Interpoles induce a commutating EMF that opposes reactance voltage.

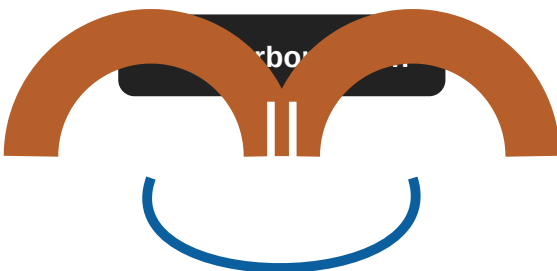
### Compensating winding

Conductors embedded in slots on the pole faces and connected in series with the armature. Their MMF opposes armature MMF under the pole arc, keeping air-gap flux nearly uniform during rapid load changes.

| Feature      | Interpoles                          | Compensating winding                               |
|--------------|-------------------------------------|----------------------------------------------------|
| Location     | Between main poles                  | In main-pole shoes                                 |
| Main purpose | Improve commutation in neutral zone | Neutralise cross-magnetising effect under pole arc |
| Connection   | Series with armature                | Series with armature                               |
| Use          | Common in medium/large machines     | Heavy-duty machines with rapidly changing load     |

### 3. Commutation

Commutation is the reversal of current in an armature coil while the brush short-circuits the two commutator segments connected to that coil. Ideally, current changes linearly from +I to -I during the commutation period and reaches its final value exactly when the coil leaves the brush.



Coil is short-circuited while brush bridges two segments.

Current must reverse from +I to -I

Inductance opposes current reversal and produces reactance voltage  $e_r = L \cdot di/dt$ . If reversal is incomplete, current interruption at the brush edge causes sparking.

#### Ideal commutation

Current reverses uniformly and reaches -I by the end of brush contact. No residual current is interrupted.

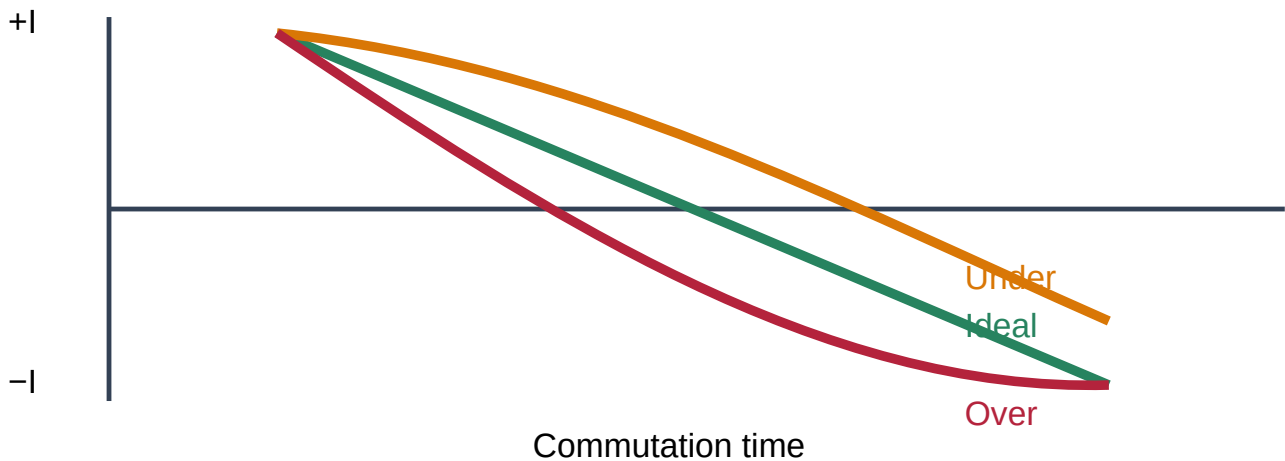
#### Under-commutation

Current reverses too slowly. At brush exit, current has not reached -I, causing sparking at the trailing edge.

#### Over-commutation

Current reverses too quickly and overshoots. Sparking may occur at the leading edge.

### Current reversal during commutation



### Methods of improving commutation

#### Carbon brushes

Higher contact resistance helps current transfer gradually from one segment to the next: resistance commutation.

#### Interpoles

Provide reversing EMF proportional to armature current: EMF commutation.

#### Correct brush position

Brushes should lie on the actual MNA, though fixed brush shift is inadequate over wide load range.

#### Shorter coil inductance

Good winding design reduces reactance voltage and improves current reversal.

## 4. Equalizer connections

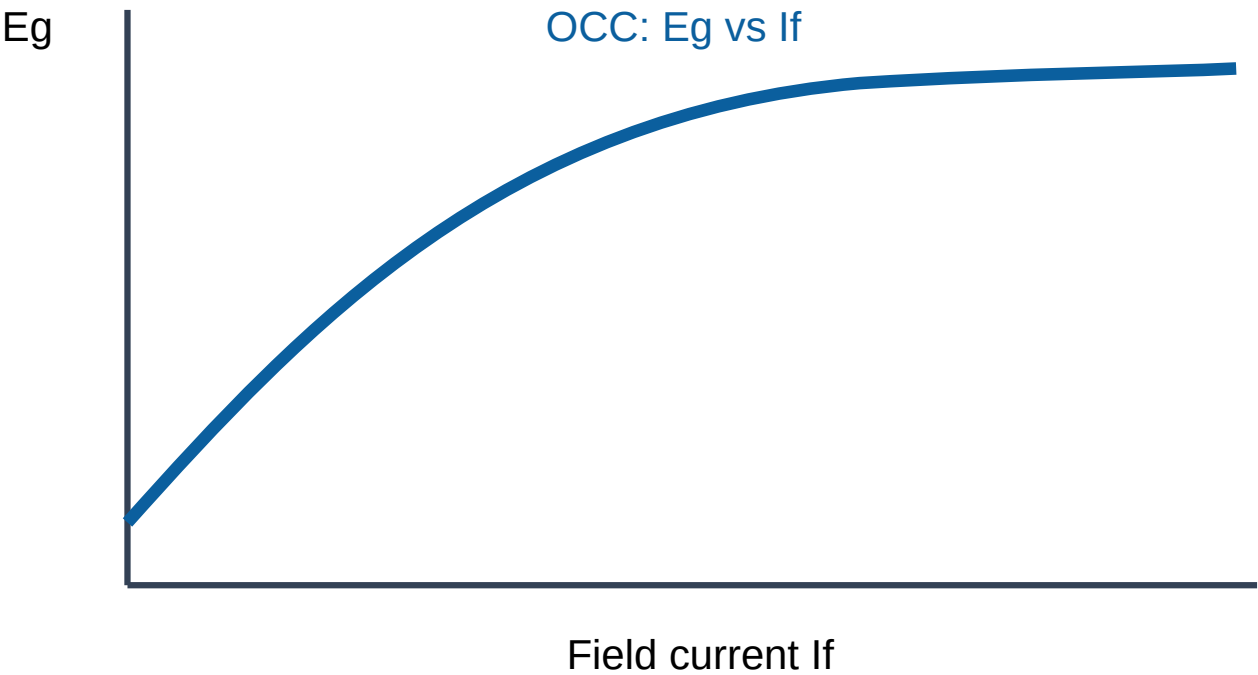
In a lap winding, each pole provides a separate parallel path. Small differences in air-gap flux can produce unequal path EMFs, causing circulating currents through brushes and commutator. Equalizer rings connect equipotential points of the winding, allowing circulating current to flow inside the armature instead of through the brushes. Wave windings normally do not require equalizer rings because each parallel path includes conductors under all poles.

## 5. Generator characteristics

A characteristic is a graph showing how voltage, current or excitation changes under specified operating conditions.

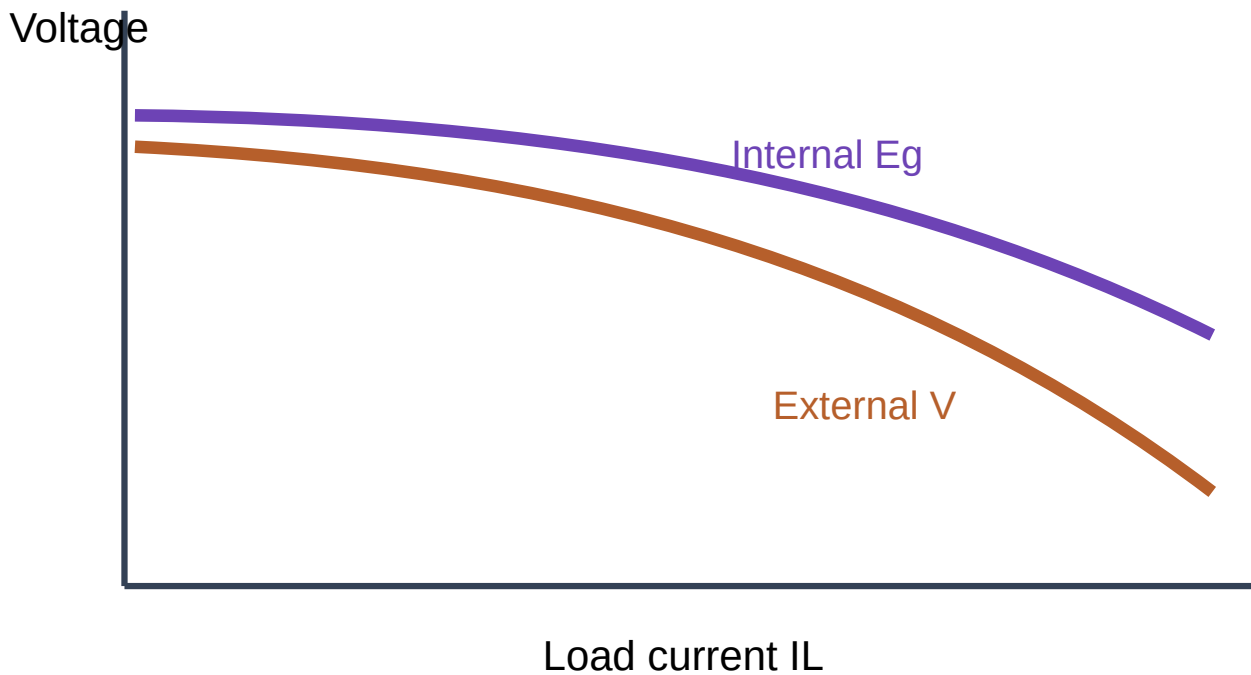
| Characteristic                    | Axes                                                             | Meaning                                                    |
|-----------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| Open-circuit characteristic (OCC) | $E_g$ versus field current $I_f$ at constant speed, no load      | Magnetisation curve; shows residual voltage and saturation |
| Internal characteristic           | Generated EMF after armature-reaction effect versus load current | Lies below ideal generated-voltage relation                |
| External characteristic           | Terminal voltage $V$ versus load current $I_L$                   | Includes armature-reaction and internal voltage drops      |

### Separately excited generator



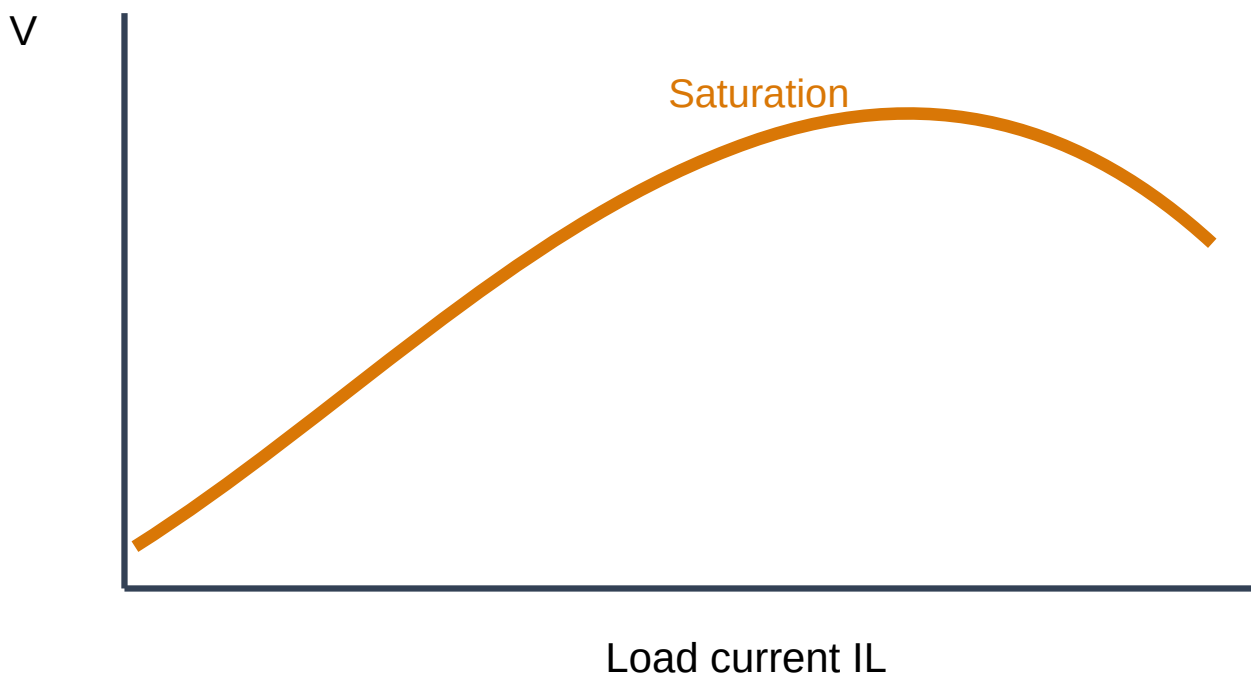
At first,  $E_g$  rises almost linearly with  $I_f$ . Magnetic saturation bends the curve.

### Shunt generator: internal and external



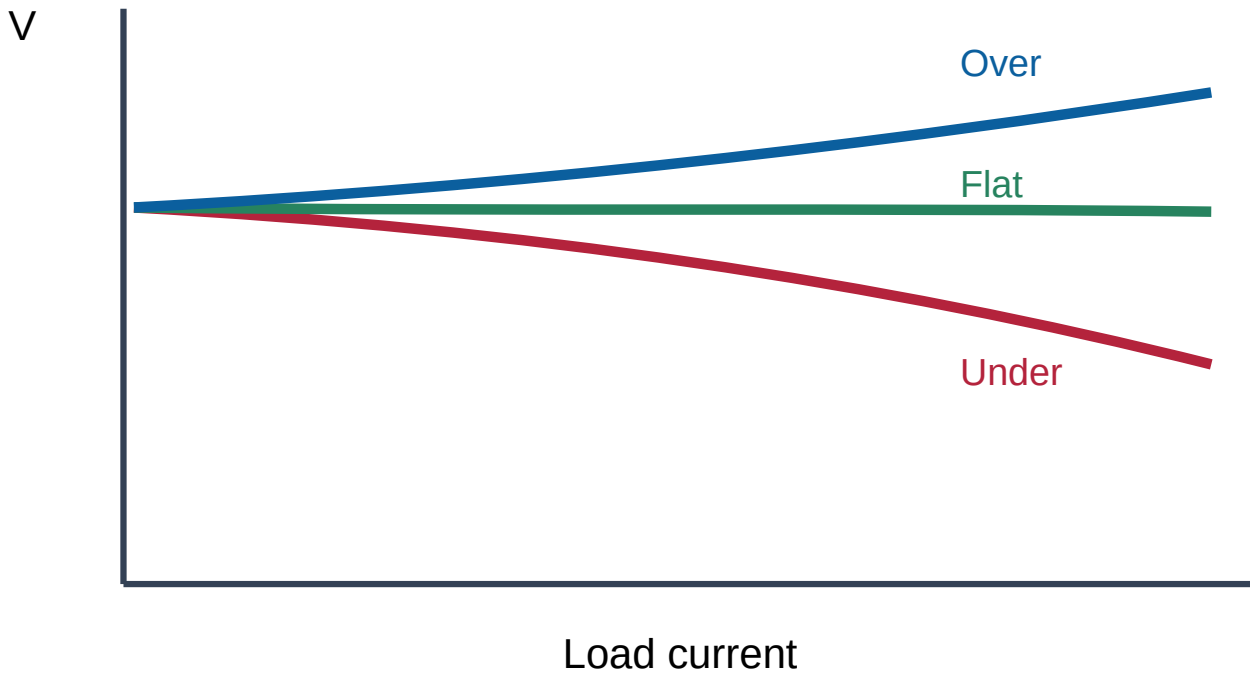
Terminal voltage falls because  $I_a R_a$  drop and armature reaction reduce voltage. Falling  $V$  also reduces shunt-field current, causing a cumulative drop.

### Series generator



Voltage initially rises with load as series-field flux increases, then saturates and may fall because of internal drops and armature reaction.

## Compound generators



## 6. Voltage build-up of a shunt generator

- 1 Residual magnetism produces a small initial EMF when the armature starts rotating.
- 2 This EMF sends current through the shunt field.
- 3 Field current strengthens flux in the same direction as residual flux.
- 4 Higher flux produces higher  $E_g$ , which produces more field current.
- 5 Voltage rises cumulatively until the field-resistance line intersects the OCC.

**Malayalam Support:** Residual magnetism  $\square\square\square\square$  voltage  $\square\square\square\square$   $\square\square\square\square\square\square\square\square$ .  
 $\square\square\square\square\square\square$  field current  $\square\square\square\square$  flux  $\square\square\square\square\square\square$ ; flux  $\square\square\square\square\square\square\square\square$  generated voltage  $\square\square\square\square\square\square$   
 $\square\square\square\square$ .  $\square\square\square\square\square$  voltage build up  $\square\square\square\square\square\square\square\square$ .

### Conditions for build-up

- Residual magnetism must exist.
- Field connection must aid residual flux.
- Field-circuit resistance must be below critical resistance.
- Speed must exceed critical speed for the chosen field resistance.
- Field circuit must be closed.



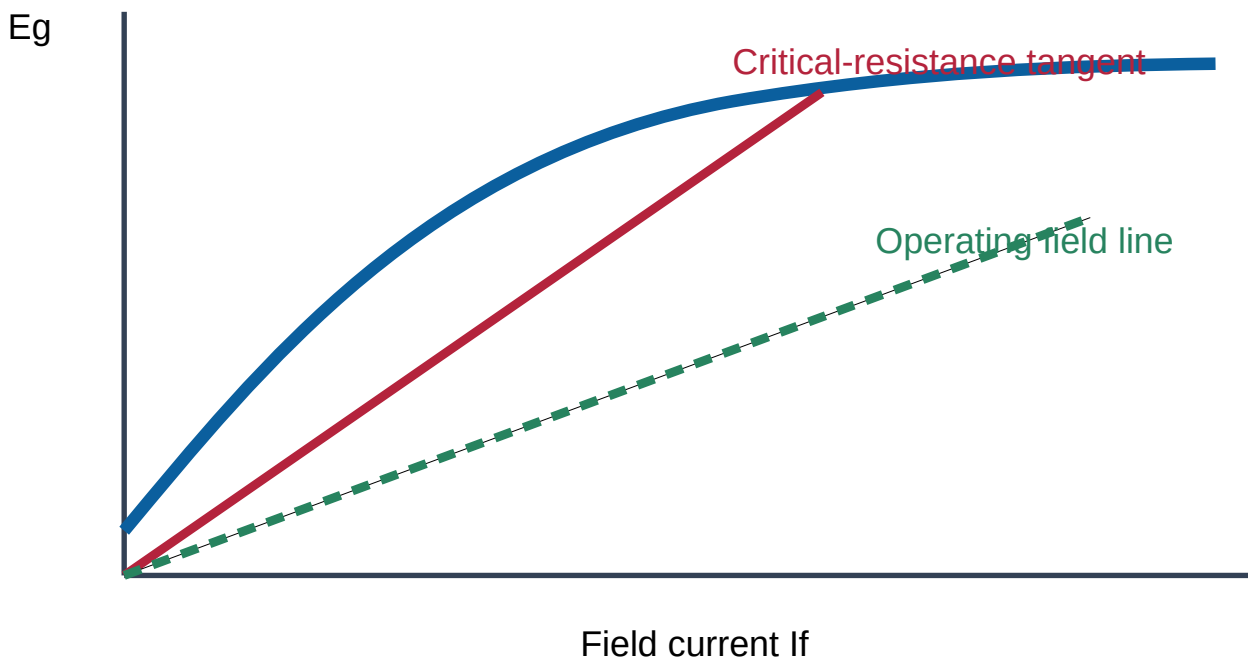
## Failure causes

- Loss of residual magnetism.
- Reversed field connection.
- Excessive field resistance.
- Speed too low.
- Open field circuit or poor brush contact.

## 7. Critical resistance and critical speed

**Critical field resistance** is the maximum shunt-field resistance at which a generator just builds voltage at a given speed. Graphically it is the slope of the line drawn from the origin tangent to the OCC. **Critical speed** is the minimum speed at which the machine builds voltage for a specified field resistance.

### Critical-resistance tangent method



### Solved numerical M2-1: Critical resistance from OCC data

**Problem:** The tangent to a shunt-generator OCC passes through  $I_f=2.0$  A,  $E_g=220$  V. Determine critical field resistance.

**Formula:**  $R_{crit} = \text{slope} = E_g/I_f$ .

**Calculation:**  $R_{crit}=220/2=110 \Omega$ .

**Final answer:** **110  $\Omega$** . The actual field-circuit resistance must be less than this at that speed.

### Solved numerical M2-2: Critical speed

**Problem:** At 1000 rpm the critical resistance is 120  $\Omega$ . Assuming the unsaturated OCC is proportional to speed, estimate critical speed for a 90  $\Omega$  field circuit.

**Relation:** Critical-resistance slope is proportional to speed, so  $N_{crit}/N_1 = R_f/R_{crit,1}$ .

**Calculation:**  $N_{crit} = 1000 \times 90 / 120 = 750$  rpm.

**Final answer:** 750 rpm.

**Assumption:** Use the linear/unsaturated portion and geometrically similar OCC.

## 8. Parallel operation of DC generators

Parallel operation improves reliability, permits maintenance without total shutdown, allows economical matching of generating capacity to load and provides reserve capacity. Before closing the parallel switch, polarity must be the same and incoming voltage must approximately equal busbar voltage.

- 1 Run the incoming generator at rated speed with its switch open.
- 2 Verify correct polarity.
- 3 Adjust excitation until incoming voltage equals bus voltage.
- 4 Close the switch.
- 5 Increase incoming-generator excitation to make it take load; adjust machines for desired sharing.

### Safety note

Wrong polarity effectively short-circuits the generators through the busbars. Never parallel without polarity and voltage checks.

Equalizer connections are essential for compound generators so unequal series-field currents do not cause one generator to take excessive load.

## 9. Application selection

| Requirement                                  | Suitable generator        | Reason                                     |
|----------------------------------------------|---------------------------|--------------------------------------------|
| Adjustable DC laboratory supply              | Separately excited        | Field can be controlled independently      |
| Nearly constant-voltage supply               | Shunt generator           | Small voltage drop with load               |
| Feeder-voltage compensation / booster        | Series generator          | Voltage increases with load current        |
| Constant terminal voltage with changing load | Flat-compounded generator | Series field offsets internal voltage drop |
| Rising voltage at remote load                | Over-compounded generator | Compensates both machine and feeder drop   |

## Module 2 expected-question practice

### Definition

#### **Define armature reaction and MNA.**

State armature-field effect and zero-EMF axis.

### Important diagram

#### **Show main, armature and resultant flux.**

Label GNA, shifted MNA and crowded pole tip.

### Comparison

#### **Interpoles versus compensating windings.**

Compare location, purpose and connection.

### Characteristic curve

#### **Draw OCC, internal and external curves.**

Label axes and explain why each curve lies below the previous one.

### Important numerical

#### **Find critical resistance or critical speed.**

Use tangent slope and proportionality with speed.

### Long answer

#### **Explain voltage build-up and failure causes.**

Include residual magnetism, polarity, resistance and speed conditions.

# DC Motors, Starters and Testing

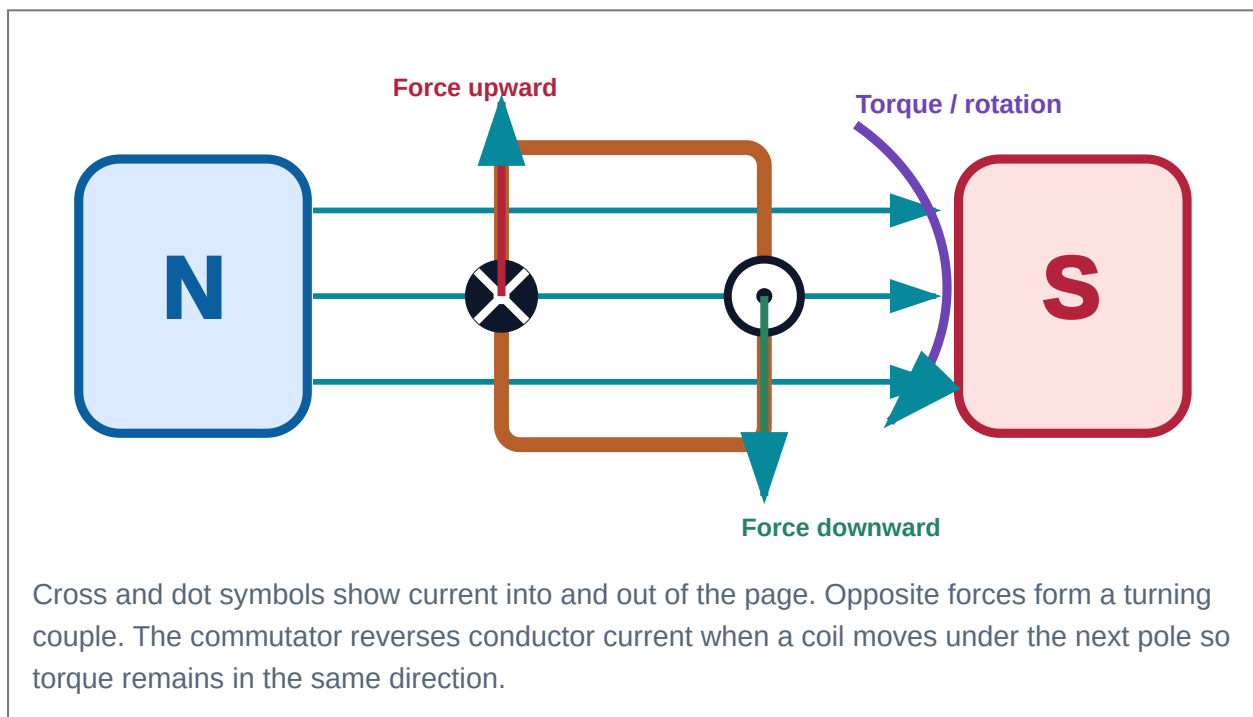
M3.01–M3.04: motor principle, back EMF, torque, performance characteristics, starters, load test and Swinburne's test.

## 1. Working principle of a DC motor

A current-carrying conductor placed in a magnetic field experiences a force. In a DC motor, forces on armature conductors form a couple and produce torque. Fleming's left-hand rule gives the force direction: first finger points along field, second finger along current and thumb along force or motion.

$$F = B I l \sin \theta$$

F in newtons; B in tesla; I in amperes; active length l in metres.



**Malayalam Support:** ഏതൊരു ചുരുക്കത്തിലും current ഏതൊരു conductor-ൽ force ഏതൊരു torque. Armature-ൽ current ഏതൊരു force-ൽ torque ഏതൊരു torque.

## 2. Back EMF and self-regulation

As the armature rotates, its conductors cut flux and generate an EMF by generator action. By Lenz's law, this induced EMF opposes the applied voltage, so it is called **back EMF**  $E_b$ . It limits armature current during normal running.

$$V = E_b + I_a R_a + V_b$$

Neglecting brush drop:  $E_b = V - I_a R_a$

$$E_b = P\Phi ZN / 60A$$

### Self-regulating action

If load increases, speed falls slightly.  $E_b$  falls,  $I_a = (V - E_b) / R_a$  rises, torque rises and the motor meets the larger load. If load decreases, speed and  $E_b$  rise, reducing  $I_a$  and torque.

**Malayalam Support:** Speed  $\propto$  back EMF;  $\propto$  armature current  $\propto$  speed. Load  $\propto$  speed  $\propto$  back EMF  $\propto$  current/torque  $\propto$ .

## 3. Why a starter is essential

At starting,  $N=0$  and therefore  $E_b=0$ . Because armature resistance is very small, connecting the motor directly to supply would cause a dangerously high current.

$$I_{a,start} = V / R_a \quad (\text{without starter})$$

$$I_{a,start} = V / (R_a + R_{st}) \quad (\text{with starter resistance})$$

### Solved numerical M3-1: Starting current and starter resistance

**Problem:** A 220 V motor has  $R_a=0.25 \Omega$ . Find direct starting current and series resistance needed to limit current to 50 A.

**Direct start:**  $I_a=220/0.25=880 \text{ A}$ .

**Required total resistance:**  $R_{total}=220/50=4.4 \Omega$ .

**Starter resistance:**  $R_{st}=4.4-0.25=4.15 \Omega$ .

**Final answer:** Direct current **880 A**; starter resistance **4.15  $\Omega$** .

## 4. Classification and connection

### Separately excited

Field supplied independently. Excellent independent flux control.

### Shunt motor

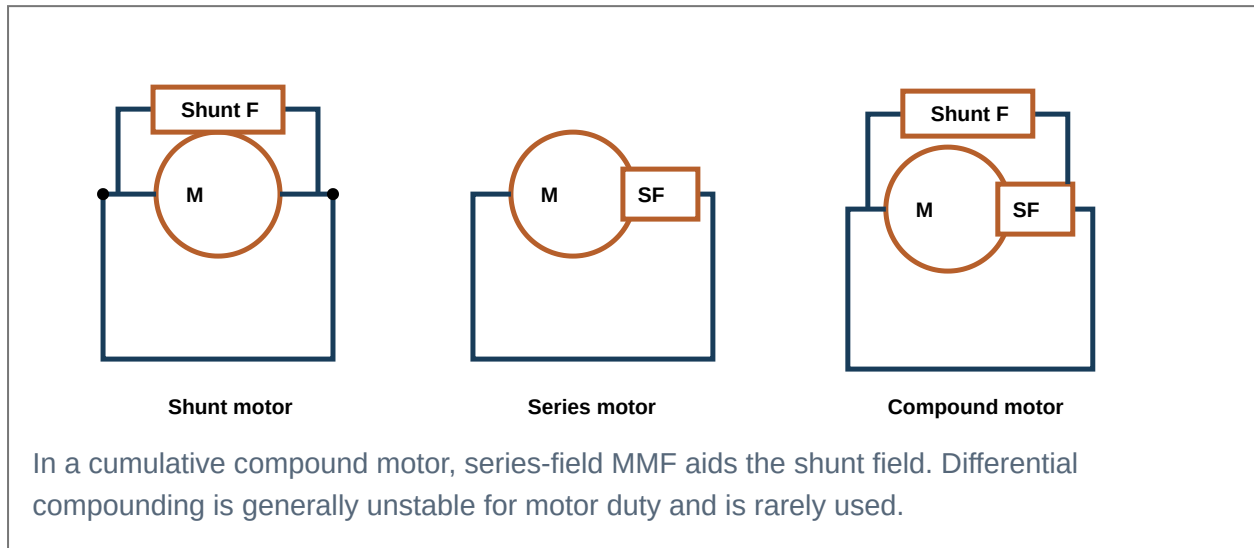
Field connected across supply. Nearly constant flux and speed.

### Series motor

Field in series with armature. Flux initially proportional to  $I_a$ ; very high starting torque.

## Compound motor

Shunt and series fields combine. Cumulative compound gives high starting torque with better speed regulation.



## 5. Mechanical power and torque

Armature electrical input is  $V I_a$ . Armature copper loss is  $I_a^2 R_a$ . The remaining power converted electromagnetically is called gross mechanical power developed.

$$P_m = E_b I_a$$

$$\omega = 2\pi N/60 \text{ rad/s}$$

$$T_a = P_m / \omega$$

### Derivation of the general torque equation

1 Mechanical power developed  $P_m = E_b I_a$ .

2 Substitute  $E_b = P \Phi Z N / (60 A)$ .

3 Angular speed  $\omega = 2\pi N / 60$ .

4 Torque  $T_a = P_m / \omega = [P \Phi Z N I_a / (60 A)] / [2\pi N / 60]$ .

5 Cancel N and simplify.

$$T_a = (PZ / 2\pi A) \Phi I_a = 0.159 (PZ / A) \Phi I_a \text{ N}\cdot\text{m}$$

$$\text{Therefore } T \propto \Phi I_a$$

**Armature torque** is the electromagnetic torque developed inside the armature. **Shaft torque** is the useful output torque after subtracting torque equivalent of iron and mechanical losses.

$$T_{sh} = P_{out}/\omega$$

### Solved numerical M3-2: Back EMF, developed power and torque

**Problem:** A 220 V shunt motor takes armature current 40 A.  $R_a=0.30\ \Omega$  and speed=1000 rpm. Neglect brush drop. Find  $E_b$ ,  $P_m$  and  $T_a$ .

**Back EMF:**  $E_b=220-40\times0.30=208\text{ V}$ .

**Developed power:**  $P_m=208\times40=8320\text{ W}$ .

**Angular speed:**  $\omega=2\pi\times1000/60=104.72\text{ rad/s}$ .

**Torque:**  $T_a=8320/104.72=79.45\text{ N}\cdot\text{m}$ .

**Final answer:**  $E_b=208\text{ V}$ ,  $P_m=8.32\text{ kW}$ ,  $T_a=79.5\text{ N}\cdot\text{m}$ .

## 6. Speed relation

$$N \propto E_b/\Phi = (V-I_a R_a)/\Phi$$

Speed can therefore be changed by varying flux, armature-circuit resistance or applied armature voltage. This relation is the foundation of Module 4.

## 7. Condition for maximum mechanical power

- 1 Ignoring brush drop,  $I_a=(V-E_b)/R_a$ .
- 2 Mechanical power developed  $P_m=E_b I_a=E_b(V-E_b)/R_a$ .
- 3 Differentiate with respect to  $E_b$ :  $dP_m/dE_b=(V-2E_b)/R_a$ .
- 4 Set derivative to zero:  $V-2E_b=0$ .

At maximum mechanical power:  $E_b = V/2$

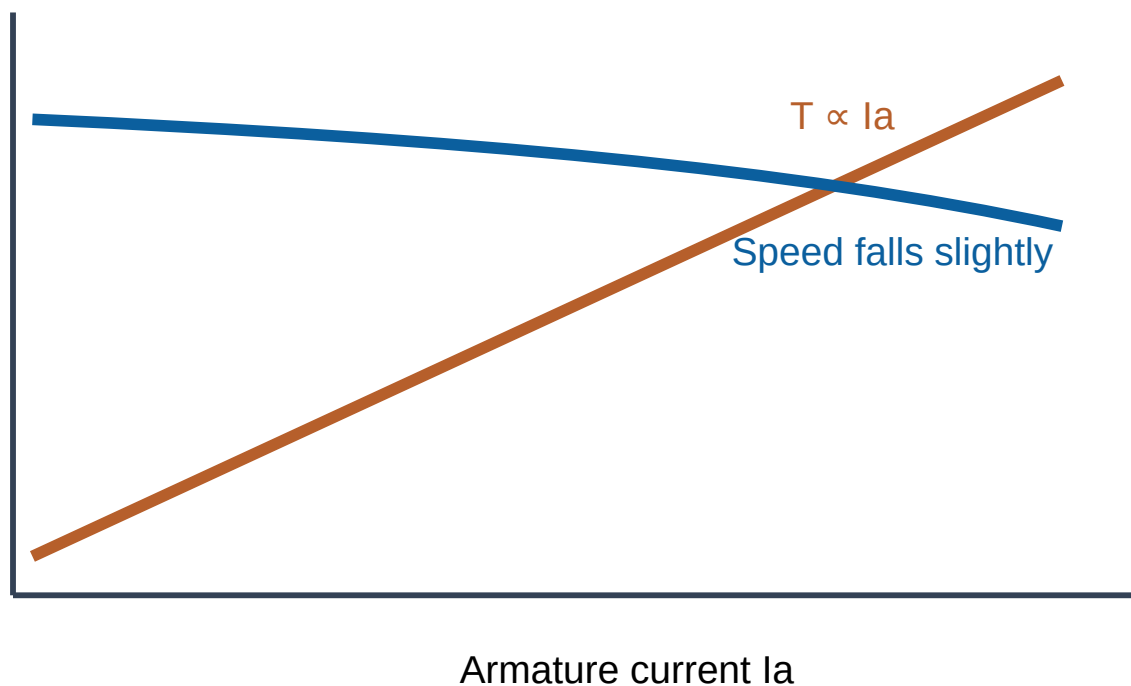
### Why this is not normal operation

At  $E_b=V/2$ ,  $I_a=V/(2R_a)$ . Armature copper loss  $I_a^2 R_a$  equals mechanical power developed, so even before rotational losses the efficiency is only 50%. Current is excessive and heating is severe.

## 8. Motor characteristics

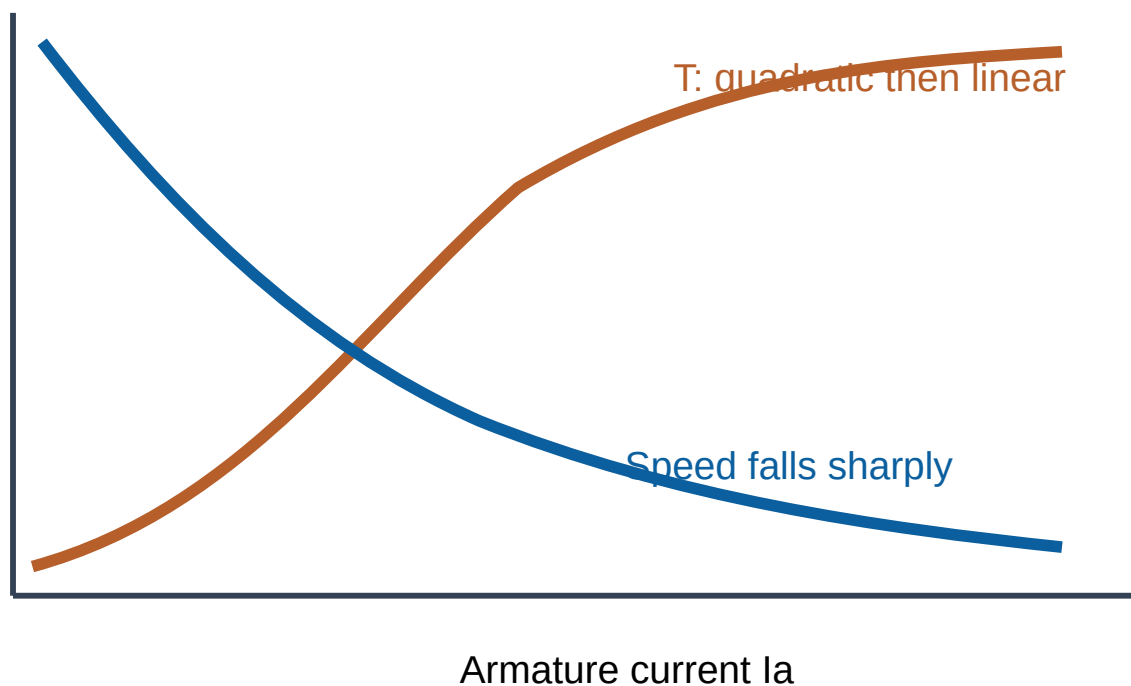
Electrical characteristic usually means torque versus armature current. Mechanical characteristic means speed versus torque. Performance characteristic includes speed versus current, efficiency versus output and other operating curves.

## Shunt motor characteristics



Flux is approximately constant: torque is proportional to  $I_a$  and speed is nearly constant.

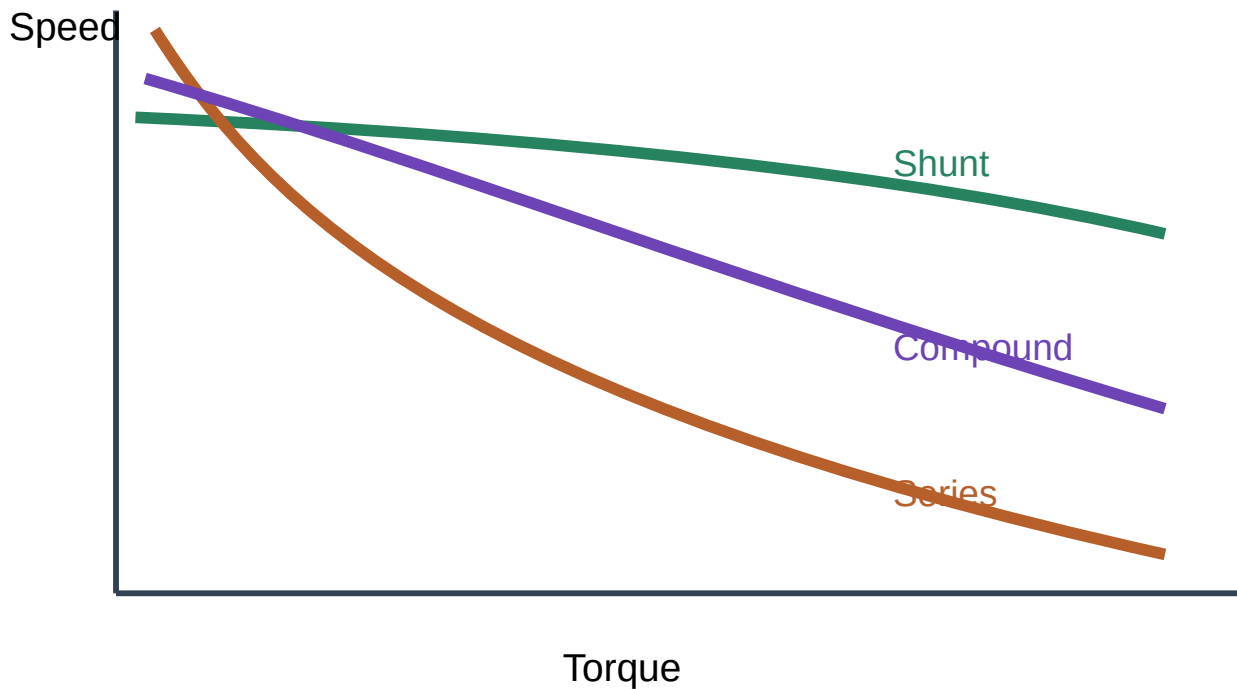
## Series motor characteristics



Before saturation,  $\Phi \propto I_a$  so  $T \propto I_a^2$ . At light load, flux is small and speed can become dangerously high; never run a series motor without load.



## Speed–torque comparison

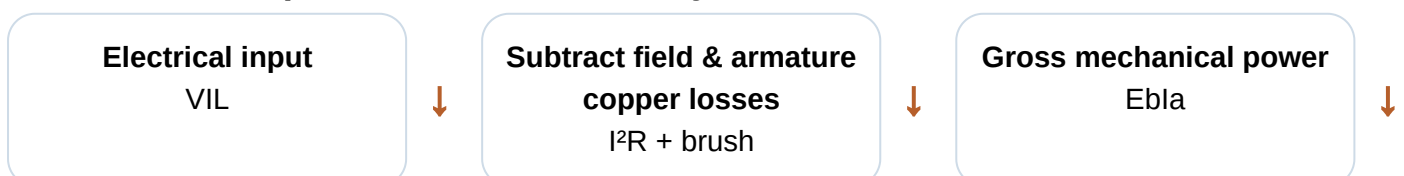


## Application selection logic

- **Nearly constant speed:** shunt motor.
- **Very high starting torque:** series motor.
- **High starting torque plus better speed regulation:** cumulative compound motor.
- **Independent wide control:** separately excited motor.

| Motor               | Starting torque | Speed regulation                | Suitable applications                                  |
|---------------------|-----------------|---------------------------------|--------------------------------------------------------|
| Shunt               | Moderate        | Good                            | Lathes, fans, blowers, pumps, machine tools            |
| Series              | Very high       | Poor; no-load dangerous         | Cranes, hoists, traction, elevators                    |
| Cumulative compound | High            | Better than series              | Conveyors, presses, rolling mills, reciprocating pumps |
| Separately excited  | Controlled      | Excellent with feedback/control | Precision drives, laboratories, variable-speed systems |

## 9. Motor losses, power flow and efficiency



Subtract rotational & stray losses



Shaft output

P<sub>out</sub>

$$\eta_m = \text{Mechanical output} / \text{Electrical input} \times 100\%$$

At maximum efficiency: variable copper loss = constant loss

Use the correct current and include field loss consistently.

### Solved numerical M3-3: Motor efficiency

**Problem:** A 220 V shunt motor takes line current 30 A. Shunt-field current is 1 A,  $R_a=0.4 \Omega$  and rotational losses are 500 W. Find shaft output and efficiency.

**Armature current:**  $I_a=30-1=29$  A.

**Input:**  $P_{in}=220 \times 30=6600$  W.

**Armature copper loss:**  $29^2 \times 0.4=336.4$  W.

**Field loss:**  $220 \times 1=220$  W.

**Shaft output:**  $6600-336.4-220-500=5543.6$  W.

**Efficiency:**  $5543.6/6600 \times 100=83.99\%$ .

**Final answer:** Output  $\approx 5.54$  kW; efficiency  $\approx 84.0\%$ .

## 10. Three-point starter

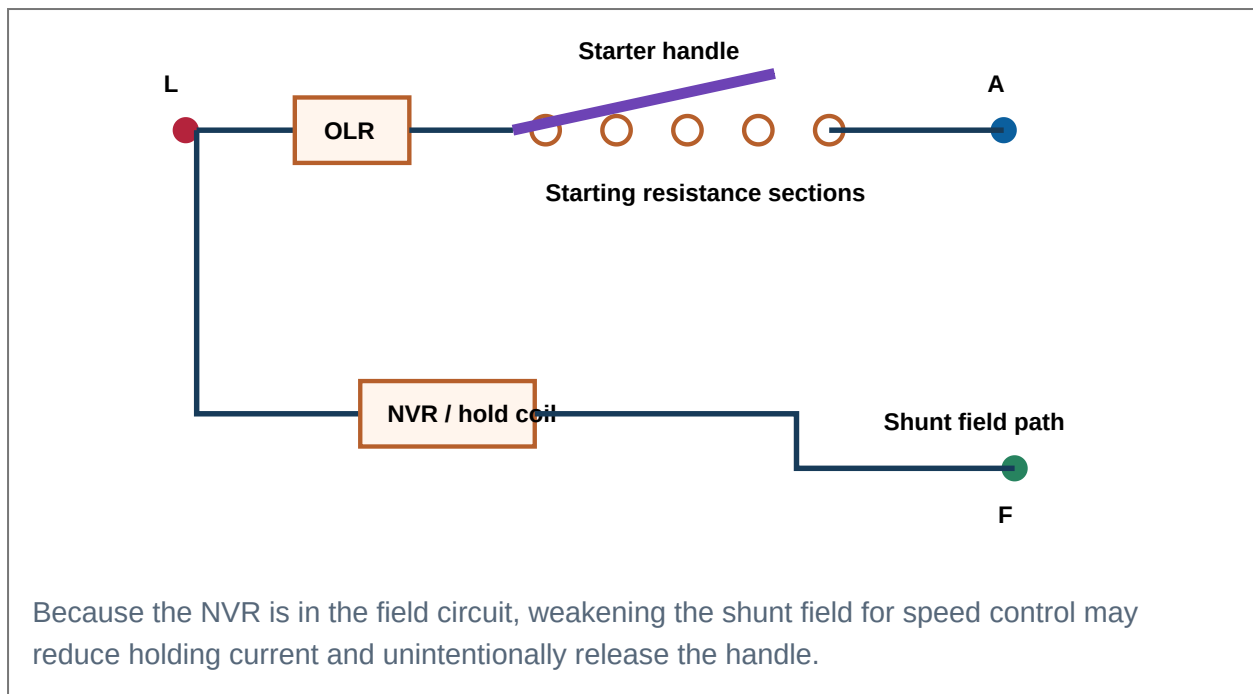
The three terminals are L (line), A (armature) and F (field). Starting resistance is divided into sections. Moving the handle from OFF toward RUN progressively removes resistance as speed and back EMF rise.

### No-volt release (NVR)

Connected in series with the shunt field. It magnetically holds the handle in RUN. Supply failure or field-circuit opening releases the handle and spring returns it to OFF.

### Overload release (OLR)

Connected in series with armature. Excess current energises it and releases the NVR mechanism, disconnecting the motor.



## 11. Four-point starter

The four terminals are L, A, F and N. The NVR coil is connected directly across the supply through a limiting resistance, independent of the shunt-field circuit. Therefore field weakening does not release the handle. However, this arrangement does not directly protect against field-circuit opening in the same way as the three-point starter.

| Feature                     | Three-point starter        | Four-point starter                      |
|-----------------------------|----------------------------|-----------------------------------------|
| Terminals                   | L, A, F                    | L, A, F, N                              |
| NVR connection              | Series with shunt field    | Across supply independently             |
| Field-control compatibility | May release at weak field  | Suitable for field weakening            |
| Field-failure protection    | Better inherent protection | Needs separate field-failure protection |

## 12. Direct load test

In a direct load test, the motor drives a brake, generator or dynamometer. Measure supply voltage, line/armature current, speed and output torque. Then calculate input, output and efficiency at different loads.

$$P_{\text{out}} = 2\pi NT/60$$

- 1 Start the motor with no load using the starter.
- 2 Record no-load readings.
- 3 Increase load gradually and record V, I, N and torque.
- 4 Compute input, output and efficiency.
- 5 Do not exceed rated current or temperature.

### Laboratory safety

Check couplings, guards, earthing and instrument ranges. Never touch rotating parts or adjust a mechanical brake without supervision.

## 13. Swinburne's test

Swinburne's test is an indirect no-load test mainly for DC shunt and compound machines. Run the motor at rated voltage and speed without mechanical load. The no-load input supplies armature copper loss, field loss and constant rotational losses. From these, efficiency at any assumed load can be predicted.

- 1 Measure supply voltage  $V$ , no-load line current  $I_0$ , field current  $I_{sh}$  and armature resistance  $R_a$ .
- 2 No-load armature current  $I_{a0} = I_0 - I_{sh}$ .
- 3 No-load input  $P_0 = VI_0$ .
- 4 Constant rotational loss  $W_c = P_0 - VI_{sh} - I_{a0}^2 R_a$  (subtract brush loss if specified).
- 5 At a chosen load, calculate armature copper loss and combine with  $W_c$  and field loss to predict efficiency.

### Advantages

- Simple and economical.
- Requires only no-load power.
- Efficiency can be predicted for motor and generator operation.
- No large loading arrangement required.

### Limitations

- Temperature rise under full load is not tested.
- Stray-load loss and armature-reaction effects are not accurately included.
- Not suitable for series motors.
- Commutation under load is not assessed.

Solved numerical M3-4: Swinburne's-test prediction

**Problem:** A 220 V shunt motor takes 5 A at no load. Field current=1 A and Ra=0.5 Ω. Predict efficiency when line current is 40 A.

**No-load armature current:**  $I_{a0}=5-1=4\text{ A}$ .

**No-load input:**  $220\times5=1100\text{ W}$ .

**Field loss:**  $220\times1=220\text{ W}$ .

**No-load armature copper:**  $4^2\times0.5=8\text{ W}$ .

**Constant rotational loss:**  $1100-220-8=872\text{ W}$ .

**At 40 A line current:**  $I_a=39\text{ A}$ ; armature copper= $39^2\times0.5=760.5\text{ W}$ .

**Total losses:**  $872+220+760.5=1852.5\text{ W}$ .

**Input:**  $220\times40=8800\text{ W}$ ; output= $6947.5\text{ W}$ .

**Efficiency:**  $6947.5/8800\times100=78.95\%$ .

**Final answer:** Predicted motor efficiency  $\approx$ **79.0%**.

| Feature                     | Direct load test                           | Swinburne's test                          |
|-----------------------------|--------------------------------------------|-------------------------------------------|
| Method                      | Actual mechanical loading                  | No-load indirect prediction               |
| Power required              | High at full load                          | Low                                       |
| Temperature and commutation | Observed under load                        | Not accurately assessed                   |
| Efficiency accuracy         | Actual at tested load                      | Predicted                                 |
| Suitable machines           | Most motors if loading equipment available | Shunt and cumulative compound; not series |

## Module 3 expected-question practice

### Very short

#### Why is a starter required?

At start  $E_b=0$ ; small  $R_a$  would cause excessive current.

### Important derivation

#### Derive $T=0.159(PZ/A)\Phi I_a$ .

Use  $P_m=E_b I_a$  and  $T_a=P_m/\omega$ .

### Important derivation

#### Derive maximum-power condition.

Differentiate  $E_b(V-E_b)/R_a$  to obtain  $E_b=V/2$  and explain low efficiency.

### Characteristic curve

#### Draw shunt and series motor curves.

Show torque-current, speed-current and speed-torque relations.

### Important diagram

#### Explain three-point and four-point starters.

Label starting resistance, handle, NVR and OLR.

### Long answer

#### Explain Swinburne's test.

Procedure, calculation, advantages, limitations and prediction numerical.

# Speed Control and Traction Motors

M4.01–M4.04: speed-control methods, DC series motors in traction, series-parallel control, shunt/bridge transitions and electrical braking.

## 1. Speed equation and control variables

$$N \propto E_b / \Phi = (V - I_a R_a) / \Phi$$

Motor speed is controlled by changing applied armature voltage  $V$ , armature-circuit resistance, or flux per pole  $\Phi$ . The practical method depends on motor type, required speed range, efficiency, load torque and cost.

### Flux control

Reduce flux to obtain speed above base speed. Efficient because field current is small, but excessive weakening reduces torque capability and may cause overspeed.

### Armature resistance control

Insert series resistance to reduce  $E_b$  and speed below base speed. Simple but inefficient; speed varies with load.

### Voltage control

Vary armature voltage while maintaining flux. Gives smooth, efficient control below base speed using controlled rectifiers, choppers or motor-generator systems.

| Method              | Speed range             | Efficiency | Load regulation         | Main limitation                            |
|---------------------|-------------------------|------------|-------------------------|--------------------------------------------|
| Flux control        | Above base speed        | High       | Good if field is stable | Reduced torque and commutation margin      |
| Armature resistance | Below base speed        | Low        | Poor                    | Large $I^2R$ loss and load-dependent speed |
| Voltage control     | Mostly below base speed | High       | Good                    | Needs variable DC source/power electronics |

## 2. Shunt-motor speed control

### Field-rheostat control

A rheostat in series with the shunt field reduces field current and flux. Since  $N \propto 1/\Phi$  approximately, speed rises. It is used above rated/base speed.

For approximately constant  $E_b$ :  $N_1/N_2 = \Phi_2/\Phi_1$

### Armature-resistance control

A resistor in series with armature increases voltage drop  $I_a R$ , reducing  $E_b$  and speed. Used for short-duration low-speed duty.

$$N_2/N_1 = [(V - I_a(R_a + R_{ext})) / \Phi_2] / [(V - I_a R_a) / \Phi_1]$$

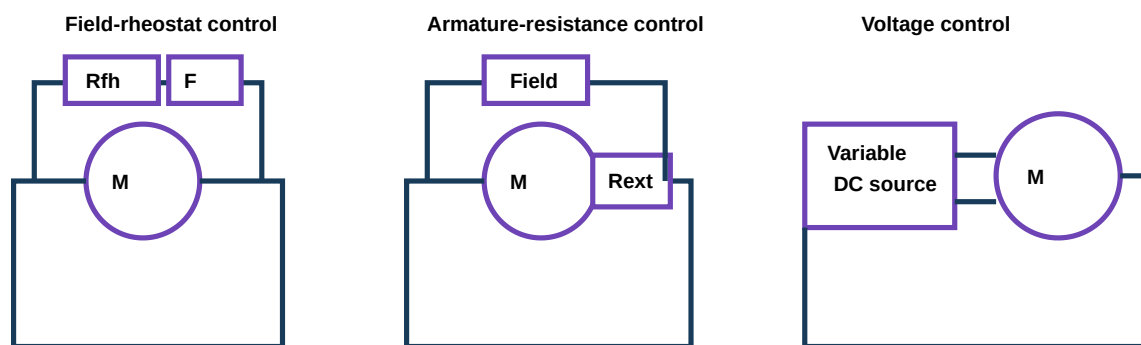
### Armature-voltage control

With constant flux, speed is nearly proportional to armature voltage. Modern chopper drives provide smooth control and regenerative capability.

$$\text{For constant } \Phi: N_1/N_2 = E_{b1}/E_{b2}$$

### Constant-torque / constant-power regions

Below base speed, voltage control with rated flux gives nearly constant torque. Above base speed, field weakening reduces flux and available torque, so output is approximately constant power within limits.



Field control changes  $\Phi$ ; armature resistance wastes power in  $R_{ext}$ ; voltage control changes armature terminal voltage with better efficiency.



### **Solved numerical M4-1: Speed after flux change**

**Problem:** A shunt motor runs at 1000 rpm with flux  $\Phi_1$ . If field flux is reduced by 20% and armature voltage drop is unchanged, find new speed.

**Given:**  $\Phi_2 = 0.8\Phi_1$ .

**Relation:**  $N_1/N_2 = \Phi_2/\Phi_1$ .

**Calculation:**  $N_2 = N_1 \times \Phi_1/\Phi_2 = 1000/0.8 = 1250$  rpm.

**Final answer:** 1250 rpm.

**Common mistake:** Speed is inversely, not directly, proportional to flux.

### **Solved numerical M4-2: Armature resistance for target speed**

**Problem:** A 220 V shunt motor runs at 1000 rpm taking  $I_a = 30$  A.  $R_a = 0.4 \Omega$ . Find external resistance for 700 rpm at the same current and flux.

**Initial Eb:**  $E_1 = 220 - 30 \times 0.4 = 208$  V.

**At constant flux:**  $E_2/E_1 = N_2/N_1 = 700/1000$ ;  $E_2 = 145.6$  V.

**Total armature resistance:**  $R_{\text{total}} = (220 - 145.6)/30 = 2.48 \Omega$ .

**External resistance:**  $R_{\text{ext}} = 2.48 - 0.4 = 2.08 \Omega$ .

**Final answer:** 2.08  $\Omega$ .

## **3. Series-motor speed control**

### **Field diverter**

A variable resistor connected in parallel with the series field diverts part of armature current, reducing field current and flux. Speed increases.

### **Armature diverter**

A resistor is connected in parallel with armature. Series-field current can exceed armature current, increasing flux and lowering speed. It is inefficient and used for special low-speed control.

### **Tapped field**

Selecting fewer series-field turns weakens flux and increases speed. Used in traction for discrete high-speed settings.

Series resistance

External armature resistance reduces motor voltage and speed. It is mainly used during starting and low-speed acceleration because energy is lost as heat.

4. Electric traction and traction-motor requirements

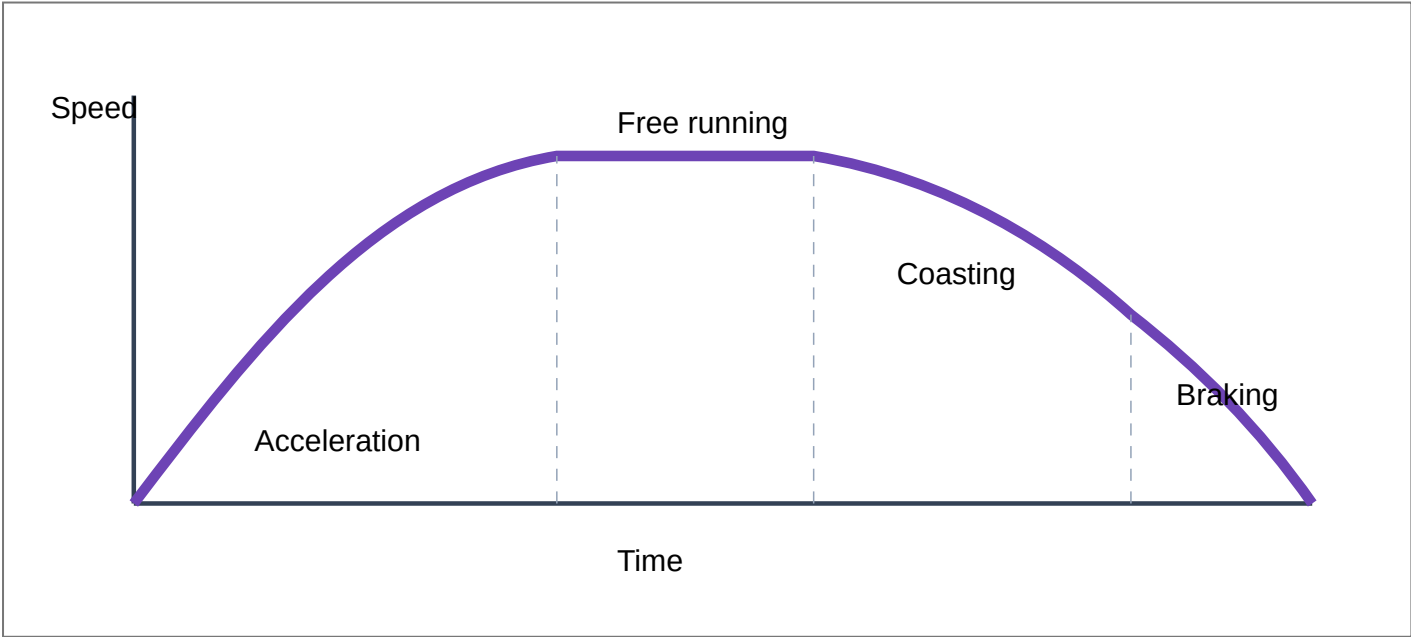
Electric traction uses electric motors to propel locomotives, multiple units, trams, trolley buses and some industrial vehicles. A traction motor repeatedly starts heavy mass, accelerates rapidly, operates over a wide speed range and must survive vibration, overload and frequent braking.

| Requirement            | Why it matters                                         | DC series-motor suitability                                  |
|------------------------|--------------------------------------------------------|--------------------------------------------------------------|
| High starting torque   | Large train mass must be accelerated from rest         | $T \propto I_a^2$ before saturation                          |
| High overload capacity | Gradients and acceleration require short-time overload | Robust series field and high torque                          |
| Wide speed control     | Starting, acceleration, cruising and grade control     | Series-parallel, resistance and field control available      |
| Electrical braking     | Reduce mechanical-brake wear and improve control       | Can operate as generator for rheostatic/regenerative braking |
| Compact and rugged     | Limited bogie space and severe vibration               | Historically well suited                                     |

Malayalam Support: DC series motor-□□ starting torque □□□□ □□□□□□□□. Train □□□□□□ □□□□ load rest-□ □□□□□ □□□□□□□□ □□□□□□□□□□ □□□ □□□□□□□□□□□□.

5. Tractive effort and speed-time curve

A typical run has acceleration, free running/coasting and braking periods. The speed-time curve helps estimate schedule speed, energy use and motor rating.



6. Series-parallel control of traction motors

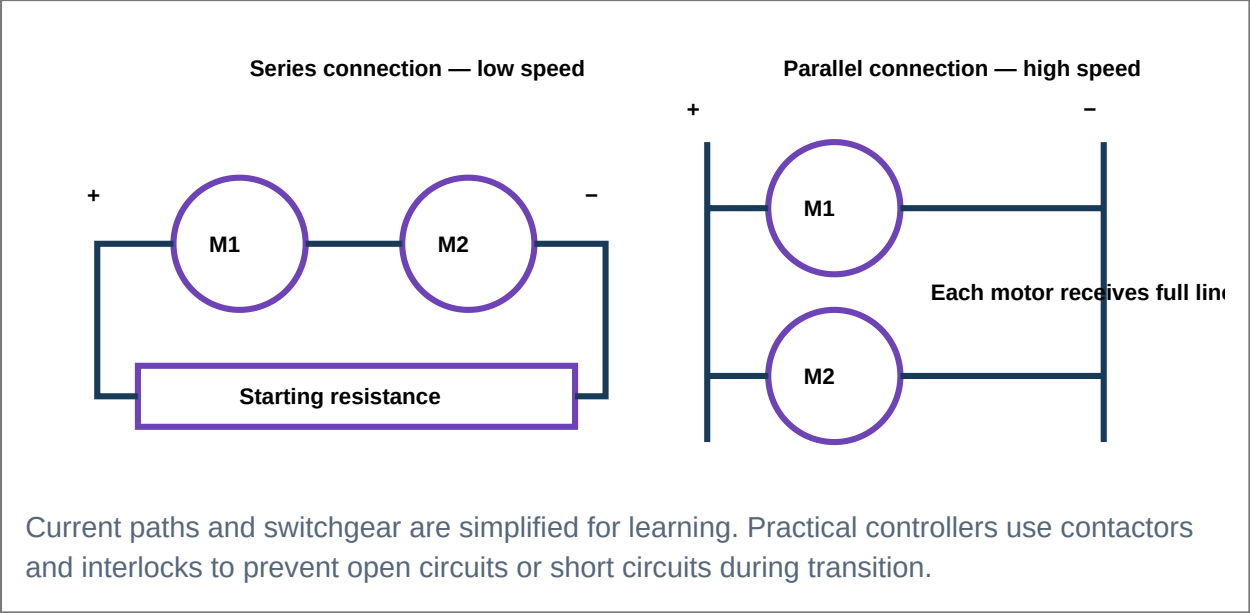
For a vehicle with two identical series motors and a fixed DC supply, starting resistances limit current. At low speed, motors are connected in series so each receives about half the supply. At higher speed, motors are

reconnected in parallel so each receives full supply. Resistance is progressively cut out in each connection.

Interactive traction-control sequence

Stage 1 — Series with full starting resistance

Motors M1 and M2 are in series. Current passes through all starting-resistance sections. Torque is controlled and each motor receives a fraction of supply voltage.



7. Transition methods

Shunt transition

During change from series to parallel, one motor is temporarily shunted by a resistor while connections are rearranged. Motor current is maintained, but some energy is lost in transition resistors.

Bridge transition

Motors and transition resistances form a bridge network. Current remains more nearly continuous and torque disturbance is lower, but switching is more complex.

| Point              | Shunt transition              | Bridge transition                             |
|--------------------|-------------------------------|-----------------------------------------------|
| Current continuity | Maintained through shunt path | Maintained through bridge paths               |
| Torque disturbance | Moderate                      | Usually lower                                 |
| Control complexity | Simpler                       | More contactors and sequencing                |
| Energy loss        | Transition resistor loss      | Transition resistor loss but smoother sharing |

Malayalam Support: Series connection-□ □□□□□ motors supply voltage □□□□□□□□□□. Parallel connection-□ □□□ motor-□□□ full voltage □□□□□□□□□□. Transition □□□□□□□ current

shunt/bridge method.

## 8. Electrical braking

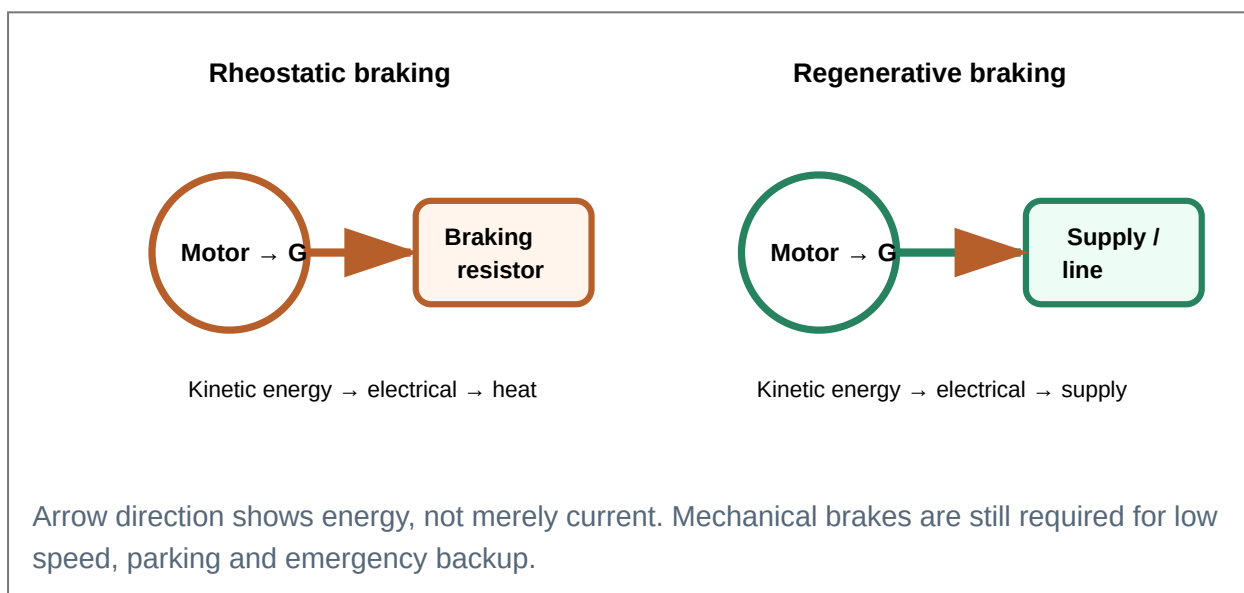
During electrical braking the motor operates as a generator and converts vehicle kinetic energy into electrical energy. The generated energy is either dissipated in resistors or returned to the supply.

## Rheostatic braking

Disconnect the motor from supply, maintain suitable field and connect armature to a braking resistor. Generated current produces torque opposite motion. Energy becomes heat in the resistor.

## Regenerative braking

Adjust excitation/speed so generated EMF exceeds supply voltage with correct polarity. Current reverses into the supply and electromagnetic torque opposes motion.



| Feature               | Rheostatic braking             | Regenerative braking                             |
|-----------------------|--------------------------------|--------------------------------------------------|
| Energy destination    | Dissipated as heat             | Returned to supply                               |
| Supply receptivity    | Not required                   | Supply must accept returned energy               |
| Control               | Relatively simple              | Requires correct voltage/polarity and protection |
| Efficiency            | Energy wasted                  | Energy recovered                                 |
| Low-speed performance | Weakens as generated EMF falls | Also weakens; mechanical brake needed            |

**Malayalam Support:** Rheostatic braking-□ generated energy resistor-□ heat □□□

□□□□□□□□□□□□□□. Regenerative braking-□ energy supply-□□□□□□ □□□□□□ □□□□□□□□□□.

### **Solved numerical M4-3: Rheostatic braking power**

**Problem:** During braking, a traction motor generates 300 V and is connected to total resistance 2.5  $\Omega$ . Neglect losses. Find braking current and electrical braking power.

**Current:**  $I = E_g / R = 300 / 2.5 = 120 \text{ A}$ .

**Power:**  $P = E_g I = 300 \times 120 = 36,000 \text{ W}$ .

**Final answer:** Braking current=**120 A**; power dissipated=**36 kW**.

### **Solved numerical M4-4: Regenerated power**

**Problem:** A motor generates 620 V while connected to a 600 V line. Total circuit resistance is 0.20  $\Omega$ . Find current returned and power delivered to the line.

**Current:**  $I = (E_g - V) / R = (620 - 600) / 0.20 = 100 \text{ A}$ .

**Line power:**  $P_{\text{line}} = VI = 600 \times 100 = 60 \text{ kW}$ .

**Generated electrical power:**  $E_g I = 62 \text{ kW}$ ; 2 kW is circuit copper loss.

**Final answer:** **100 A** returned; **60 kW** delivered to line.

## Module 4 expected-question practice

### Formula

**State the DC-motor speed equation.**

$$N \propto (V - I_a R_a) / \Phi.$$

### Important numerical

**Find speed after flux, resistance or voltage change.**

Compare back EMFs and fluxes; state constant-current/flux assumptions.

### Application

**Why is a DC series motor suitable for traction?**

High starting torque, overload capability, wide control and electrical braking.

### Important diagram

**Draw series and parallel motor connections.**

Show starting resistances and current paths.

### Comparison

**Shunt transition versus bridge transition.**

Compare current continuity, torque disturbance and switching complexity.

### Long answer

**Explain rheostatic and regenerative braking.**

Include energy flow, operating condition, advantages and limitations.

## FORMULA BANK

# Module-wise equations, units, conditions and common mistakes

Use SI units unless a problem states otherwise. Convert mWb to Wb, kW to W and rpm to rad/s where required.

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## Module 1 — Generator formulas

### Generated EMF

$$E_g = P\Phi ZN / (60A)$$

**Units:**  $E_g$  V,  $\Phi$  Wb, N rpm.

**Condition:** Average generated DC value at brushes.

**Mistake:** Wrong parallel paths. Lap  $A=P$ ; wave  $A=2$ .

**Mini example:**  $P=4$ ,  $\Phi=.02$ ,  $Z=500$ ,  $N=1200$ , wave  $\rightarrow E_g=400$  V.

### Terminal voltage

$$V = E_g - I_a R_a - V_b$$

**Units:** V, A,  $\Omega$ .

**Condition:** Generator delivering current.

**Mistake:** Brush drop must be included only if given/assumed.

### Shunt current relation

$$I_{sh} = V / R_{sh}; I_a = I_L + I_{sh}$$

**Condition:** Shunt generator.

**Mistake:** Armature current is greater than load current.

### Series generator

$$I_a = I_L; V = E_g - I_a (R_a + R_{se})$$

**Condition:** Series field and armature carry same current.

### Generator efficiency

$$\eta_g = P_{out} / P_{in} \times 100\%$$

**Output:** VIL. **Input:** output + all losses.

**Mistake:** Do not divide output by generated power if rotational losses exist.

### Maximum efficiency

Variable copper loss = constant loss

**Typical:**  $I^2R = W_c$ .

**Condition:** Approximately constant voltage and constant rotational/field losses.

## Module 2 — Characteristic rules

### Critical resistance

$R_{crit} = \text{slope of tangent from origin to OCC} = E/I_f$

Maximum field-circuit resistance for build-up at a given speed. Read E and  $I_f$  from the tangent, not an arbitrary OCC point.

### Critical speed

$N_{crit}/N_1 \approx R_f/R_{crit,1}$

Use only when OCCs scale proportionally with speed in the unsaturated region. Critical speed is the minimum speed for the given field resistance.

### Voltage regulation

$\% \text{ regulation} = (V_{NL} - V_{FL})/V_{FL} \times 100\%$

Use when asked. Keep the denominator convention stated by the problem or institution.

### Commutation reactance voltage

$e_r = L \cdot di/dt$

Shows why rapid current reversal in an inductive coil causes sparking. This is a concept relation, not normally a detailed numerical topic here.



## Module 3 — Motor formulas

### Voltage and back EMF

$$V = E_b + I_a R_a; E_b = V - I_a R_a$$

Neglect brush drop unless stated. Back EMF is less than supply during motoring.

### Back-EMF equation

$$E_b = P \Phi Z N / (60 A)$$

Same machine constant as generator EMF. Use motor speed and flux.

### Speed relation

$$N \propto E_b / \Phi = (V - I_a R_a) / \Phi$$

For comparisons, use ratios and state which quantities are constant.

### Torque

$$T = 0.159 (P Z / A) \Phi I_a$$

**Unit:** N·m. **Mistake:**  $\Phi$  must be in webers, not mWb.

### Mechanical power developed

$$P_m = E_b I_a$$

This is gross converted mechanical power before rotational losses.

### Torque from power

$$T = P / \omega = 60 P / (2 \pi N) = 9.55 P / N$$

If P is watts and N rpm, T is N·m. With P in kW:  $T = 9550 P(\text{kW}) / N$ .

### Starting current

$$I_{a,start} = V / (R_a + R_{st})$$

At standstill  $E_b = 0$ . Solve  $R_{st} = V / I_{limit} - R_a$ .

### Motor efficiency

$$\eta_m = P_{shaft} / P_{electrical\ input} \times 100\%$$

Line input includes field current in a shunt motor.

### Maximum mechanical power

$$E_b = V / 2$$

Idealised armature-circuit result. It gives excessive current and  $\leq 50\%$  conversion efficiency before rotational losses.

## Module 4 — Speed-control and braking formulas

### General speed ratio

$$N_2 / N_1 = [(V_2 - I_2 R_2) / (V_1 - I_1 R_1)] \cdot (\Phi_1 / \Phi_2)$$

R includes armature plus external series resistance. Use consistent currents.

### Constant flux

$$N_1 / N_2 = E_{b1} / E_{b2}$$

Suitable for armature-voltage or resistance-control comparisons.

### Approximately constant $E_b$

$$N_1 / N_2 = \Phi_2 / \Phi_1$$

Used for field-control estimates when armature drop is small.

### Rheostatic braking

$$I = E_g / R_{\text{total}}; P = E_g I = I^2 R$$

Generated energy is dissipated in resistors.

### Regenerative braking

$$I = (E_g - V) / R_{\text{total}}, \text{ when } E_g > V$$

Current returns to the supply; line power approximately  $VI$ .

### Mechanical output / braking torque

$$T = 9.55 P / N$$

Use the converted mechanical/electrical power consistent with the requested torque and account for losses if supplied.

## DC MACHINE CONSTRUCTION BANK

# Parts, magnetic path and maintenance insight

### Yoke

Outer frame, mechanical support and flux return. Inspect for cracks, loose feet and earthing.

### Pole core

Supports field coil and carries flux. Tight pole bolts preserve air-gap uniformity.

### Pole shoe

Spreads flux, lowers reluctance and supports field coil.

### Field winding

Creates main MMF. Shunt coils use fine wire/many turns; series coils use thick wire/few turns.

### **Armature core**

Laminated steel cylinder with slots. Laminations reduce eddy-current loss.

### **Armature winding**

Coils in slots connected to commutator segments. Insulation must withstand thermal and mechanical stress.

### **Commutator**

Copper segments insulated by mica. Surface should be concentric, smooth and properly undercut.

### **Brushes**

Carbon contacts collect/supply current. Correct grade, pressure and neutral position prevent sparking.

### **Shaft**

Supports armature and transmits torque. Check alignment and key condition.

### **Bearings**

Maintain rotation and air gap. Lubrication, noise and temperature indicate condition.

### **Interpoles**

Series-connected auxiliary poles that improve current reversal under brushes.

### **Compensating winding**

Pole-face conductors that cancel armature MMF under load.

### Machine insight

Many brush and commutation faults are actually caused by mechanical problems: eccentric commutator, vibration, incorrect spring pressure, dirty surface, poor alignment or overloaded bearings.

## CIRCUIT AND CONNECTION DIAGRAM BANK

# What to show in every examination diagram

| Diagram                      | Essential labels and current path                                            | Key equation / note                     |
|------------------------------|------------------------------------------------------------------------------|-----------------------------------------|
| Separately excited generator | Independent field supply, armature $E_g/R_a$ , load, $I_a$ and $I_L$         | Field current independent of load       |
| Shunt generator              | Shunt field across terminals; $I_a$ splits into $I_L$ and $I_{sh}$           | $I_a = I_L + I_{sh}$                    |
| Series generator             | Series field in load path                                                    | $I_a = I_L$                             |
| Compound generator           | Both shunt and series fields; show long/short-shunt connection               | Mark aiding/opposing polarity           |
| Shunt motor                  | Armature and shunt field across supply; $I_a$ and $I_{sh}$ branches          | $I_L = I_a + I_{sh}$                    |
| Series motor                 | Series field and armature in one path                                        | Never run unloaded                      |
| Compound motor               | Both fields; mark cumulative direction                                       | Series field aids shunt field           |
| Three-point starter          | L,A,F; starting resistance, handle, NVR, OLR                                 | NVR in field circuit                    |
| Four-point starter           | L,A,F,N; NVR across supply                                                   | Independent of field current            |
| Load test                    | Supply, starter, ammeter, voltmeter, motor, brake/dynamometer and tachometer | $P_{out} = 2\pi NT/60$                  |
| Swinburne test               | No-load shunt motor, V/A meters, field rheostat                              | Find constant losses from no-load input |
| Flux control                 | Field rheostat in shunt-field circuit                                        | Weak field $\rightarrow$ high speed     |
| Armature control             | Series resistor or variable armature supply                                  | Lower $E_b \rightarrow$ lower speed     |
| Traction series/parallel     | Two motors, starting resistors, contactors and current path                  | Series low speed; parallel high speed   |
| Rheostatic braking           | Motor reconnected as generator to resistor                                   | Energy $\rightarrow$ heat               |
| Regenerative braking         | Generated EMF opposing and exceeding supply                                  | Energy $\rightarrow$ supply             |

### Diagram-marking rule

Use standard symbols, show terminal names, draw current arrows, state switch position and add a one-sentence working principle below the diagram.

## CHARACTERISTIC CURVE BANK

# Axes, shape and interpretation

| Curve                    | Horizontal axis     | Vertical axis                         | Important interpretation                                                                            |
|--------------------------|---------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------|
| Generator OCC            | Field current $I_f$ | Generated EMF $E_g$                   | Residual voltage, linear region, knee and saturation                                                |
| Internal generator       | Load current        | Generated EMF after armature reaction | Below ideal/no-reaction curve                                                                       |
| External generator       | Load current        | Terminal voltage $V$                  | Below internal curve by resistive drops                                                             |
| Series generator         | Load current        | Terminal voltage                      | Rises, saturates, may fall                                                                          |
| Compound generator       | Load current        | Terminal voltage                      | Under, flat or over compounded                                                                      |
| Shunt motor T–<br>$I_a$  | Armature current    | Torque                                | Nearly straight line through origin                                                                 |
| Shunt motor N–<br>$I_a$  | Armature current    | Speed                                 | Slight droop with load                                                                              |
| Series motor T–<br>$I_a$ | Armature current    | Torque                                | Quadratic before saturation; linear after                                                           |
| Series motor N–<br>$I_a$ | Armature current    | Speed                                 | Very high at light load; falls with current                                                         |
| Speed-control curve      | Torque/load         | Speed                                 | Armature resistance gives drooping lower-speed curves; voltage control gives nearly parallel curves |

### Characteristic insight

A complete graph answer needs labeled axes, operating condition (constant speed/voltage/flux), important regions such as saturation, and a short physical reason for the curve shape.

## POWER-FLOW AND LOSSES BANK

# Where power goes in generators and motors

### Generator

Mechanical input → subtract iron/mechanical losses → gross electrical power  $E_g I_a$  → subtract armature/field/brush losses → terminal output  $V I_L$ .

# Motor

Electrical input  $VIL$  → subtract field/armature/brush losses → gross mechanical power  $E_{b1a}$  → subtract rotational/stray losses → shaft output.

| Loss                           | Cause                                              | Variation                               | Reduction                                     |
|--------------------------------|----------------------------------------------------|-----------------------------------------|-----------------------------------------------|
| Armature copper<br>$I_a^2 R_a$ | Armature resistance                                | Variable with current squared           | Low-resistance copper, cooling                |
| Field copper                   | Field-winding resistance                           | Shunt nearly constant; series variable  | Correct conductor size                        |
| Hysteresis                     | Repeated magnetisation of armature iron            | Depends on frequency and flux density   | Silicon steel                                 |
| Eddy current                   | Induced currents in iron                           | Depends on flux and speed               | Thin insulated laminations                    |
| Mechanical                     | Bearing/brush friction and windage                 | Approximately constant near rated speed | Lubrication, alignment and ventilation design |
| Stray load                     | Leakage, armature reaction, current redistribution | Increases with load                     | Good magnetic and winding design              |

## MOTOR SELECTION AND APPLICATION BANK

# Select by torque, regulation, overload and control

| Application                | Load requirement                                 | Recommended motor               | Technical justification                                 |
|----------------------------|--------------------------------------------------|---------------------------------|---------------------------------------------------------|
| Lathe / machine tool       | Nearly constant speed, adjustable range          | Shunt or separately excited     | Good regulation and convenient field/voltage control    |
| Fan / blower               | Low starting torque, smooth speed                | Shunt                           | Stable speed and moderate starting duty                 |
| Centrifugal pump           | Nearly constant speed                            | Shunt                           | Good regulation                                         |
| Conveyor                   | Higher starting torque, varying load             | Cumulative compound             | Good starting torque with better regulation than series |
| Crane / hoist              | Very high starting torque and overload           | Series                          | Torque rises strongly with current                      |
| Elevator                   | High starting torque and controlled acceleration | Series or compound with control | Strong starting performance                             |
| Rolling mill               | Shock load and high overload                     | Cumulative compound             | Torque support with acceptable speed change             |
| Electric traction          | Very high starting torque, wide control, braking | DC series (traditional)         | Excellent torque and series-parallel control            |
| Printing machine           | Controlled nearly constant speed                 | Shunt / separately excited      | Good regulation                                         |
| Reciprocating pump / press | Fluctuating heavy torque                         | Cumulative compound             | Handles peak torque without extreme speed rise          |

### Selection method

First identify starting torque and overload requirement, then speed regulation, duty cycle, control range, braking and supply. Do not select a motor only from power rating.

### TRACTION SYSTEM BANK



# Series-parallel stages, transition and braking

| Stage / method        | Connection or energy path                 | Purpose                                          | Main limitation                        |
|-----------------------|-------------------------------------------|--------------------------------------------------|----------------------------------------|
| Series + resistance   | Motors in series with starting resistors  | Limit starting current                           | High resistor loss                     |
| Full series           | Motors in series, resistors removed       | Efficient low/medium speed                       | Each motor receives about half voltage |
| Shunt transition      | Temporary shunt paths during reconnection | Maintain motor current                           | Torque disturbance and resistor loss   |
| Bridge transition     | Bridge network balances current paths     | Smoother series-to-parallel change               | More complex contactor sequence        |
| Parallel + resistance | Motors parallel with resistance           | Limit current after transition                   | Temporary loss                         |
| Full parallel         | Each motor across supply                  | High-speed operation                             | Higher voltage per motor               |
| Rheostatic braking    | Generated energy to resistor              | Electrical braking independent of receptive line | Energy wasted as heat                  |
| Regenerative braking  | Generated energy to supply                | Energy recovery and reduced brake wear           | Needs receptive supply and control     |

## SOLVED NUMERICAL EXAMPLES

# Additional worked problems across all modules

These examples complement the in-module problems and follow the required examination format.

### S1. Flux per pole from generator data

**Problem:** A 4-pole simplex wave generator has  $Z=720$ ,  $N=500$  rpm and  $E_g=240$  V. Find flux per pole.

**Formula:**  $\Phi=60AE_g/(PZN)$ , with  $A=2$ .

**Substitution:**  $\Phi=(60\times2\times240)/(4\times720\times500)=0.02$  Wb.

**Answer:** 0.02 Wb = 20 mWb per pole.

## S2. Number of armature conductors

**Problem:** A 6-pole lap generator must generate 250 V at 750 rpm with  $\Phi=30$  mWb. Find Z.

**Formula:**  $Z=60AE_g/(P\Phi N)$ ,  $A=P=6$ .

**Calculation:**  $Z=(60 \times 6 \times 250)/(6 \times 0.03 \times 750)=666.7$ .

**Answer:** Use approximately **668 conductors** (an even practical integer).

## S3. Lap versus wave generated voltage

**Problem:** A 4-pole armature has  $Z=480$ ,  $\Phi=25$  mWb and  $N=1000$  rpm. Find  $E_g$  for simplex lap and wave windings.

**Lap:**  $A=4$ ;  $E_g=(4 \times 0.025 \times 480 \times 1000)/(60 \times 4)=200$  V.

**Wave:**  $A=2$ ;  $E_g=(4 \times 0.025 \times 480 \times 1000)/(60 \times 2)=400$  V.

**Answer:** Lap=**200 V**; wave=**400 V**. The wave winding doubles series conductors per path.

## S4. Shunt-generator terminal voltage and currents

**Problem:**  $E_g=250$  V,  $R_a=0.15$   $\Omega$ ,  $R_{sh}=100$   $\Omega$  and load current  $I_L=80$  A. Neglect brush drop. Find V,  $I_{sh}$  and  $I_a$ .

Because  $I_{sh}=V/R_{sh}$  and  $I_a=I_L+I_{sh}$ ,  $V=E_g-R_a(I_L+V/R_{sh})$ .

$V[1+R_a/R_{sh}]=E_g-R_a I_L=250-12=238$ .

$V=238/(1.0015)=237.64$  V;  $I_{sh}=2.376$  A;  $I_a=82.376$  A.

**Answer:**  $V \approx 237.6$  V,  $I_{sh} \approx 2.38$  A,  $I_a \approx 82.38$  A.

## S5. Generator efficiency from losses

**Problem:** A 230 V generator supplies 60 A. Armature copper loss=500 W, field loss=250 W, iron+mechanical loss=700 W and stray loss=150 W. Find efficiency.

**Output:**  $230 \times 60=13,800$  W. **Total loss:** 1600 W. **Input:** 15,400 W.

**Efficiency:**  $13,800/15,400 \times 100=89.61\%$ .

**Answer:** **89.6%**.

### S6. Voltage regulation

**Problem:** A shunt generator has no-load voltage 250 V and full-load voltage 230 V. Find regulation using full-load voltage as denominator.

$$\% \text{ regulation} = (250 - 230) / 230 \times 100 = 8.70\%.$$

**Answer: 8.7%.**

### S7. Motor speed from machine constants

**Problem:** A 4-pole wave motor has  $Z=600$ ,  $\Phi=20$  mWb and  $E_b=240$  V. Find speed.

**Formula:**  $N = 60AE_b / (P\Phi Z)$ ,  $A=2$ .

$$N = (60 \times 2 \times 240) / (4 \times 0.02 \times 600) = 600 \text{ rpm}.$$

**Answer: 600 rpm.**

### S8. Shaft torque

**Problem:** A motor delivers 12 kW at 900 rpm. Find shaft torque.

**Formula:**  $T_{sh} = 9550P(\text{kW})/N$ .

$$T_{sh} = 9550 \times 12 / 900 = 127.33 \text{ N}\cdot\text{m}.$$

**Answer: 127.3 N·m.**

### S9. General torque equation

**Problem:** A 6-pole lap motor has  $Z=720$ ,  $\Phi=25$  mWb and  $I_a=50$  A. Find armature torque.

$$A=P=6. \quad T = 0.159(PZ/A)\Phi I_a = 0.159 \times 720 \times 0.025 \times 50 = 143.1 \text{ N}\cdot\text{m}.$$

**Answer: 143 N·m.**

### S10. Maximum-power operating point

**Problem:** A 240 V motor has  $R_a=0.4 \Omega$ . Under ideal assumptions find  $E_b$ ,  $I_a$  and developed mechanical power at maximum power.

$$E_b = V/2 = 120 \text{ V}. \quad I_a = (240 - 120) / 0.4 = 300 \text{ A}.$$

$$P_m = E_b I_a = 120 \times 300 = 36 \text{ kW}. \quad \text{Armature copper loss is also } 300^2 \times 0.4 = 36 \text{ kW}.$$

**Answer:  $E_b=120$  V,  $I_a=300$  A,  $P_m=36$  kW.** This is thermally unacceptable normal operation.

### S11. New speed after voltage change

**Problem:** A separately excited motor runs at 1200 rpm with  $E_b=220$  V. Flux is constant. Find speed when  $E_b$  is reduced to 165 V.

$$N_2/N_1 = E_{b2}/E_{b1} = 165/220 = 0.75. \quad N_2 = 900 \text{ rpm.}$$

**Answer: 900 rpm.**

### S12. Speed with simultaneous voltage and flux change

**Problem:** Initially  $E_{b1}=200$  V,  $\Phi_1=30$  mWb and  $N_1=1000$  rpm. Find  $N_2$  if  $E_{b2}=180$  V and  $\Phi_2=24$  mWb.

$$N_2/N_1 = (E_{b2}/E_{b1})(\Phi_1/\Phi_2) = (180/200)(30/24) = 1.125.$$

**Answer: 1125 rpm.**

## EXPECTED QUESTIONS

# Module-wise examination practice patterns

These are high-priority practice patterns inferred from the official syllabus. They are not guaranteed examination questions.

## Module 1

- **One word:** Mechanical rectifier, winding for high current.
- **Short answer:** Purpose of laminations, commutator and brushes.
- **Diagram:** Labeled DC machine and simple-loop generator.
- **Comparison:** Lap versus wave; shunt versus series generator.
- **Derivation:** EMF equation; maximum efficiency.
- **Numerical:** Eg,  $N$ ,  $\Phi$ ,  $Z$ , terminal voltage and efficiency.

## Module 2

- **Definition:** Armature reaction, commutation, critical resistance/speed.
- **Diagram:** MNA shift, interpoles, commutation.
- **Curve:** OCC, internal, external and compound characteristics.
- **Long answer:** Voltage build-up and conditions.
- **Application:** Select generator for a stated use.
- **Numerical:** Critical resistance, critical speed and voltage regulation.

## Module 3

- **Short answer:** Back EMF and starter necessity.
- **Derivation:** Torque equation and maximum mechanical power.
- **Curve:** Shunt/series motor characteristics.
- **Diagram:** Three-point/four-point starter.
- **Testing:** Load test and Swinburne's test.
- **Numerical:**  $E_b$ ,  $I_a$ , speed, torque, output, efficiency and starter resistance.

## Module 4

- **Formula:** Speed relation and ratios.
- **Comparison:** Flux, armature and voltage control.
- **Application:** Why series motor is used for traction.
- **Diagram:** Series-parallel connection and transition.
- **Long answer:** Rheostatic and regenerative braking.
- **Numerical:** New speed, external resistance and braking power.

### PRACTICE SECTION

## Objective, diagram, curve, numerical and selection exercises

### A. Fill in the blanks

1. The outer frame that completes the magnetic path is the \_\_\_\_\_.
2. In a simplex wave winding, the number of parallel paths is \_\_\_\_\_.
3. The maximum field resistance for shunt-generator voltage build-up is called \_\_\_\_\_.
4. At motor starting, back EMF is \_\_\_\_\_.
5. For a DC motor, electromagnetic torque is proportional to \_\_\_\_\_.
6. The coil that holds a three-point starter handle in RUN is the \_\_\_\_\_ release.
7. In rheostatic braking, energy is dissipated in a \_\_\_\_\_.
8. During full-parallel traction operation, each motor receives \_\_\_\_\_ supply voltage.

### B. True or false

1. A wave winding is preferred for low-voltage, high-current machines.
2. Interpoles are connected in series with armature current.
3. A series motor may safely run at no load.
4. At maximum mechanical power,  $E_b = V/2$  under simplified assumptions.
5. Field weakening increases shunt-motor speed.
6. Regenerative braking returns electrical energy to a receptive supply.

## C. Multiple-choice questions

| No. | Question                                                                          | Options                                                                            |
|-----|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1   | Mechanical rectification is performed by                                          | A yoke • B armature core • C commutator • D pole shoe                              |
| 2   | A simplex lap winding has parallel paths equal to                                 | A 2 • B P • C Z • D N                                                              |
| 3   | Critical resistance is obtained graphically from                                  | A external curve • B OCC tangent • C speed-torque curve • D efficiency curve       |
| 4   | Before saturation, series-motor torque is approximately proportional to           | A $I_a$ • B $I_a^2$ • C $1/I_a$ • D speed                                          |
| 5   | The starter suitable for field-weakening control without NVR release is           | A two-point • B three-point • C four-point • D direct switch                       |
| 6   | At full series traction position, two identical motors each receive approximately | A full line voltage • B half line voltage • C zero voltage • D double line voltage |
| 7   | Electrical energy is recovered in                                                 | A plugging • B rheostatic braking • C regenerative braking • D mechanical braking  |
| 8   | Swinburne's test is mainly suitable for                                           | A series motors • B shunt machines • C induction motors • D transformers only      |

## D. Part-identification and connection questions

1. From the construction diagram, identify the part that spreads pole flux and supports the field coil.
2. Identify the part made of copper segments separated by mica.
3. Draw a shunt generator and mark  $I_a$ ,  $I_L$  and  $I_{sh}$ .
4. Draw a three-point starter and label L, A, F, NVR and OLR.
5. Draw two traction motors in series and then in parallel, showing current path.

## E. Characteristic-curve interpretation

1. Why does the external characteristic of a shunt generator fall more steeply than its internal characteristic?
2. What does the knee of an OCC indicate?
3. Why does a series-motor speed-current curve rise sharply at light current?
4. Which compound-generator curve is preferred for feeder voltage-drop compensation?

## F. Formula and numerical practice

1. A 4-pole wave generator has  $Z=800$ ,  $\Phi=20$  mWb and  $N=600$  rpm. Find  $E_g$ .
2. A 6-pole lap generator has  $E_g=300$  V,  $Z=900$  and  $N=1000$  rpm. Find  $\Phi$ .
3. A 250 V shunt generator delivers 50 A;  $I_{sh}=2$  A and  $R_a=0.2$   $\Omega$ . Find  $E_g$ , neglecting brush drop.
4. A 220 V motor has  $E_b=205$  V and  $R_a=0.3$   $\Omega$ . Find  $I_a$ .
5. A motor develops 10 kW at 750 rpm. Find torque.
6. A shunt motor runs at 900 rpm. Flux is reduced to 75% with unchanged  $E_b$ . Find new speed.
7. A 240 V motor with  $R_a=0.4$   $\Omega$  must limit starting current to 60 A. Find starter resistance.
8. A braking generator produces 360 V across 3  $\Omega$ . Find current and power.

## G. Motor-selection scenarios

1. Select a motor for a lathe requiring nearly constant speed and explain.

2. Select a motor for a crane lifting heavy loads from rest.
3. Select a motor for a conveyor with fluctuating load but acceptable speed regulation.
4. Select a speed-control method for efficient smooth control below base speed.
5. Select a braking method where recovered energy can be returned to the line.

## H. Self-assessment checklist

- ✓ I can label all main DC-machine parts and trace the magnetic path.
- ✓ I can derive  $E_g = P\Phi ZN/60A$  and choose the correct parallel paths.
- ✓ I can calculate generator voltage, losses and efficiency.
- ✓ I can explain armature reaction, commutation and voltage build-up.
- ✓ I can draw and interpret generator characteristic curves.
- ✓ I can derive motor torque and maximum-power condition.
- ✓ I can select motors from performance requirements.
- ✓ I can explain starters and Swinburne's test.
- ✓ I can solve flux, resistance and voltage speed-control problems.
- ✓ I can trace series-parallel traction stages and braking energy flow.

### QUICK REVISION

## Compact last-day revision sheet

### Construction

Yoke, poles, field coils, laminated armature, winding, commutator, brushes, shaft, bearings, interpoles and compensating winding.

### Generator principle

Flux cutting induces EMF. Right-hand rule gives direction. Commutator makes brush output unidirectional.

### EMF equation

$E_g = P\Phi ZN/60A$ . Lap  $A=P$ ; wave  $A=2$ .

### Generator voltage

$V = E_g - I_a R_a - V_b$ . Shunt:  $I_a = I_L + I_{sh}$ .

## Losses

Copper, iron, mechanical and stray. Maximum efficiency when variable copper loss equals constant loss.

## Windings

Lap: high current/low voltage. Wave: high voltage/lower current.

## Armature reaction

Armature MMF distorts flux and shifts MNA. Compensating winding cancels pole-face distortion.

## Commutation

Current reversal under brush. Carbon brush and interpole improve commutation.

## Characteristics

OCC:  $E_g - I_f$ . Internal includes armature reaction. External includes voltage drops.

## Build-up

Residual magnetism, correct field connection,  $R_f < R_{crit}$  and  $speed > N_{crit}$ .

## Motor principle

Force on current-carrying conductor. Left-hand rule. Commutator keeps torque unidirectional.

## Back EMF

$E_b = V - I_a R_a$ . It limits current and makes the motor self-regulating.



## Torque

$T = 0.159(PZ/A)\Phi I_a$ .  $P_m = E_b I_a$ .  $T_{sh} = 9550P(\text{kW})/N$ .

## Maximum power

$E_b = V/2$  under simplified assumptions; not a normal efficient operating point.

## Characteristics

Shunt: nearly constant speed. Series: very high starting torque and dangerous no-load speed.

## Starters

Three-point NVR in field circuit; four-point NVR across supply. OLR protects against excess current.

## Testing

Load test gives actual performance. Swinburne predicts efficiency from no-load losses.

## Speed control

Flux control above base; armature resistance below base but inefficient; voltage control efficient below base.

## Traction

Series motors for high starting torque. Motors series at low speed and parallel at high speed.

## Braking

Rheostatic: energy to resistor. Regenerative: energy returned to supply.

## Common examination mistakes

Using  $mWb$  without conversion; wrong  $A$  for lap/wave; mixing generator and motor voltage signs; using line current instead of armature current; omitting field loss; drawing curves without axes; saying a series motor can run unloaded; confusing maximum power with maximum efficiency; claiming window.print automatically downloads a PDF; omitting energy-flow arrows in braking diagrams.

### Important derivations

Generator EMF equation, generator maximum-efficiency condition, motor torque equation and motor maximum-mechanical-power condition.

### Important diagrams

DC machine, simple-loop generator, armature reaction, commutation, OCC/critical tangent, three-point and four-point starters, motor characteristics, series-parallel traction and both electrical-braking circuits.

## FRESH MODEL QUESTION PAPER

# DC Machines & Traction Motors — Course 3032

Time: 3 Hours | Maximum Marks: 75

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## PART A

Answer all questions. Each carries 1 mark. ( $9 \times 1 = 9$ )

1. Name the part of a DC machine that acts as a mechanical rectifier.
2. Write the number of parallel paths in a simplex wave winding.
3. State one essential condition for voltage build-up of a shunt generator.
4. Name the auxiliary pole used to improve commutation.
5. Write the motor voltage equation, neglecting brush drop.
6. Which DC motor has the highest starting torque?
7. Name the release that protects a starter against excessive armature current.
8. Which speed-control method is normally used above base speed in a shunt motor?
9. Name the braking method that returns energy to the supply.

## PART B

Answer any eight questions. Each carries 3 marks. ( $8 \times 3 = 24$ )

1. State the functions of yoke, pole shoe and commutator.
2. Compare simplex lap and simplex wave windings.
3. A 4-pole wave generator has 600 conductors, flux 20 mWb/pole and speed 1000 rpm. Calculate generated EMF.
4. Define armature reaction and list two effects.
5. Distinguish critical field resistance and critical speed.
6. Explain why back EMF makes a DC motor self-regulating.
7. Write the torque equation and define each symbol.
8. Compare three-point and four-point starters.
9. State the procedure of Swinburne's test.
10. Compare rheostatic and regenerative braking.

## PART C

Answer one alternative from each group. Each group carries 7 marks. ( $6 \times 7 = 42$ )

1. **(A)** Explain the construction and working principle of a DC generator with a labeled diagram and derive the EMF equation. **OR (B)** A 6-pole lap generator has  $Z=720$ ,  $\Phi=30$  mWb,  $N=800$  rpm,  $R_a=0.2 \Omega$  and supplies 100 A. Field current is 2 A. Find  $E_g$ ,  $I_a$ , terminal voltage and output power. Neglect brush drop.
2. **(A)** Classify DC-generator losses, draw the power-flow diagram and derive the maximum-efficiency condition. **OR (B)** Explain armature windings and compare lap and wave windings with applications.
3. **(A)** Explain armature reaction, interpoles and compensating windings with diagrams. **OR (B)** Explain shunt-generator OCC, critical resistance, critical speed and voltage build-up conditions.
4. **(A)** Derive the general torque equation of a DC motor and explain shunt and series motor characteristics. **OR (B)** A 220 V shunt motor takes line current 45 A;  $I_{sh}=1.5$  A,  $R_a=0.25 \Omega$  and speed=950 rpm. Rotational losses are 600 W. Find  $E_b$ , developed torque, shaft output and efficiency.
5. **(A)** Explain construction and working of three-point and four-point starters. **OR (B)** Explain the direct load test and Swinburne's test, including advantages and limitations.

6. (A) Explain shunt- and series-motor speed-control methods with diagrams and comparison. **OR**  
(B) Explain series-parallel traction control, shunt/bridge transition, rheostatic braking and regenerative braking.

### COMPLETE MODEL ANSWERS

## Full solutions for all questions and both OR alternatives

### Part A

| No. | Answer                                                                                                                 |
|-----|------------------------------------------------------------------------------------------------------------------------|
| 1   | Commutator                                                                                                             |
| 2   | 2                                                                                                                      |
| 3   | Any one: residual magnetism, correct field polarity, field resistance below critical value, speed above critical speed |
| 4   | Interpole / commutating pole                                                                                           |
| 5   | $V = E_b + I_a R_a$                                                                                                    |
| 6   | DC series motor                                                                                                        |
| 7   | Overload release                                                                                                       |
| 8   | Flux or field-control method                                                                                           |
| 9   | Regenerative braking                                                                                                   |

### Part B

#### B1. Functions

**Yoke:** provides mechanical support and a low-reluctance return path for flux. **Pole shoe:** spreads flux over the armature, reduces air-gap reluctance and supports the field coil. **Commutator:** acts as a mechanical rectifier in a generator and reverses armature-coil current at the correct instant in a motor.

#### B2. Lap versus wave

Simplex lap has  $A=P$ , usually  $P$  brush sets, is suited to low-voltage high-current duty and needs equalizer rings. Simplex wave has  $A=2$ , usually two brush sets, is suited to high-voltage lower-current duty and normally does not need equalizer rings.

#### B3. Generated EMF

$P=4$ ,  $A=2$ ,  $Z=600$ ,  $\Phi=.02$  Wb,  $N=1000$  rpm.

$$E_g = \frac{P\Phi ZN}{60A} = \frac{(4 \times .02 \times 600 \times 1000)}{(60 \times 2)} = 400 \text{ V.}$$

**Answer: 400 V.**

#### **B4. Armature reaction**

Armature reaction is the effect of armature-current MMF on the main field. It distorts flux, shifts MNA, reduces average flux/generated voltage and can cause brush sparking and local pole-tip heating.

#### **B5. Critical resistance and speed**

Critical resistance is the maximum shunt-field resistance at which a generator just builds voltage at a specified speed; graphically it is the tangent slope from the OCC origin. Critical speed is the minimum speed at which voltage builds for a given field resistance.

#### **B6. Self-regulation by back EMF**

When load rises, speed and  $E_b$  fall slightly. Since  $I_a = (V - E_b)/R_a$ , armature current and torque rise to meet the load. When load falls, speed/ $E_b$  rise, current falls and torque reduces.

#### **B7. Torque equation**

$T = 0.159(PZ/A)\Phi I_a$  N·m, where P is poles, Z total armature conductors, A parallel paths,  $\Phi$  flux per pole in Wb and  $I_a$  armature current in A.

#### **B8. Three-point versus four-point starter**

Three-point terminals are L, A, F and its NVR is in series with the shunt field; field weakening may release the handle, but field failure is sensed. Four-point terminals are L, A, F, N and its NVR is connected across the supply independently, so field weakening does not release it; separate field-failure protection is desirable.

#### **B9. Swinburne's procedure**

Run the shunt motor at rated voltage without mechanical load. Measure V, no-load line current  $I_0$ , field current  $I_{sh}$  and  $R_a$ . Find  $I_{a0} = I_0 - I_{sh}$ . Constant rotational loss is  $VI_0 - VI_{sh} - I_{a0}^2 R_a$ , with brush loss included if specified. Use this constant loss plus copper losses at an assumed load to predict efficiency.

#### **B10. Braking comparison**

In rheostatic braking the motor generates into a resistor and energy becomes heat; it does not require a receptive supply but wastes energy. In regenerative braking  $E_g$  exceeds supply voltage with correct polarity, current returns to the supply and energy is recovered; more control and a receptive supply are required.

## Part C — Group 1

### C1(A). Construction, principle and EMF derivation

**Construction:** The yoke supports the machine and carries return flux. Main poles and field windings create stationary flux. The laminated armature core carries conductors and rotates on the shaft. Copper commutator segments insulated by mica connect armature coils to stationary carbon brushes. Bearings support rotation; interpoles and compensating windings improve loaded operation.

**Working principle:** A prime mover rotates armature conductors through magnetic flux. By Faraday's law an EMF is induced. Fleming's right-hand rule gives direction. The EMF in each coil reverses every half-turn, while the commutator reverses coil connection to the brushes, producing unidirectional terminal voltage.

**Derivation:** Flux cut by one conductor per revolution =  $P\Phi$  Wb. Time per revolution =  $60/N$  s. EMF per conductor =  $P\Phi N/60$  V. Conductors in series per path =  $Z/A$ . Therefore:

$$E_g = (P\Phi N/60)(Z/A) = P\Phi ZN/60A$$

For simplex lap  $A=P$ ; for simplex wave  $A=2$ . A complete diagram should label yoke, pole, field coil, armature, commutator, brushes and shaft.

### C1(B). Generator numerical

**Given:**  $P=6$ , lap  $A=6$ ,  $Z=720$ ,  $\Phi=.03$  Wb,  $N=800$  rpm,  $R_a=.2\ \Omega$ ,  $I_L=100$  A,  $I_{sh}=2$  A.

**Generated EMF:**  $E_g = (6 \times .03 \times 720 \times 800) / (60 \times 6) = 288$  V.

**Armature current:**  $I_a = I_L + I_{sh} = 102$  A.

**Terminal voltage:**  $V = E_g - I_a R_a = 288 - 102 \times .2 = 267.6$  V.

**Output power:**  $P_{out} = V I_L = 267.6 \times 100 = 26,760$  W = 26.76 kW.

**Final:**  $E_g=288$  V,  $I_a=102$  A,  $V=267.6$  V, output=26.76 kW.

## Part C — Group 2

### C2(A). Losses, power flow and maximum efficiency

**Copper losses:**  $I_a^2 R_a$ , series-field  $I^2 R$ , shunt-field  $V^2/R_{sh}$  and brush-contact loss. **Iron losses:** hysteresis and eddy currents. **Mechanical losses:** bearing/brush friction and windage. **Stray-load losses:** additional loaded effects.

**Power flow:** Mechanical input → subtract iron and mechanical losses → gross electrical power  $E_g I_a$  → subtract armature, brush and field losses → terminal output  $V I_L$ .

Let output =  $V I$ , variable loss =  $I^2 R$  and constant loss =  $W_c$ .  $\eta = VI / (VI + I^2 R + W_c)$ . Differentiating with respect to  $I$  and setting  $d\eta/dI = 0$  gives  $I^2 R = W_c$ .

Maximum efficiency occurs when variable copper loss equals constant loss.

Assume nearly constant voltage and constant rotational/field losses.

### C2(B). Armature windings

An armature winding consists of coils placed in slots and connected to commutator segments. Each turn has two active coil sides. Parallel paths determine current and voltage rating.

**Lap winding:** Coils connect to nearby commutator segments and lap back. Simplex  $A=P$ . It carries high current at lower voltage, usually uses  $P$  brush sets and needs equalizer rings.

**Wave winding:** Coils progress around the armature in a wave, connecting conductors under similar pole polarities. Simplex  $A=2$ . It gives higher voltage at lower current, uses two main brush sets and generally needs no equalizer rings.

**Applications:** Lap—heavy-current generators, electroplating and welding supplies. Wave—higher-voltage lighting or general DC generators.

## Part C — Group 3

### C3(A). Armature reaction, interpoles and compensation

Armature current produces armature MMF. Its field combines with the main field, crowds flux at one pole tip, weakens it at the other, shifts MNA and can reduce average flux. Effects include sparking, voltage reduction, local heating and poor commutation.

**Interpoles:** Small poles between main poles, series-connected with armature. Their commutating EMF opposes reactance voltage and allows brushes to remain near geometrical neutral.

**Compensating winding:** Conductors embedded in pole shoes and series-connected with armature. Their MMF opposes armature MMF under the pole arc, preventing flux distortion during rapid load change.

A complete diagram must show main poles, armature, shifted MNA, interpoles and pole-face compensation conductors.

### C3(B). OCC, critical values and voltage build-up

The OCC plots  $E_g$  against  $I_f$  at constant speed and no load. It begins at residual voltage, rises approximately linearly and bends at magnetic saturation.

Critical resistance is the maximum field resistance for build-up at a given speed and equals the slope of the tangent from origin to OCC. Critical speed is the minimum speed for build-up at a fixed field resistance.

Build-up sequence: residual magnetism creates initial  $E_g$  → field current flows → field MMF aids residual flux →  $E_g$  rises → operation settles at the field-line/OCC intersection.

Conditions: residual magnetism, correct field polarity, closed field circuit,  $R_f$  below critical resistance and speed above critical speed.

## Part C — Group 4

### C4(A). Torque derivation and characteristics

$P_m = E_b I_a$  and  $E_b = P \Phi Z N / (60 A)$ . Angular speed  $\omega = 2\pi N / 60$ . Therefore:

$$T = P_m / \omega = [P \Phi Z N I_a / (60 A)] / [2\pi N / 60] = P Z \Phi I_a / (2\pi A) = 0.159 (P Z / A) \Phi I_a$$

Thus  $T \propto \Phi I_a$ .

**Shunt motor:**  $\Phi$  approximately constant, so  $T \propto I_a$ . Speed is nearly constant and drops slightly with load. Suitable for lathes, fans and pumps.

**Series motor:** Before saturation  $\Phi \propto I_a$ , so  $T \propto I_a^2$ ; after saturation  $T \propto I_a$ . Speed is inversely related to flux and becomes dangerously high at light load. Suitable for cranes, hoists and traction.

### C4(B). Motor numerical

**Given:**  $V = 220 \text{ V}$ ,  $I_L = 45 \text{ A}$ ,  $I_{sh} = 1.5 \text{ A}$ ,  $R_a = .25 \Omega$ ,  $N = 950 \text{ rpm}$ , rotational loss =  $600 \text{ W}$ .

**Armature current:**  $I_a = 45 - 1.5 = 43.5 \text{ A}$ .

**Back EMF:**  $E_b = 220 - 43.5 \times .25 = 209.125 \text{ V}$ .

**Developed power:**  $P_m = E_b I_a = 209.125 \times 43.5 = 9096.94 \text{ W}$ .

**Developed torque:**  $T_a = 9550 \times 9.09694 / 950 = 91.44 \text{ N}\cdot\text{m}$ .

**Shaft output:**  $9096.94 - 600 = 8496.94 \text{ W}$ .

**Input:**  $220 \times 45 = 9900 \text{ W}$ .

**Efficiency:**  $8496.94 / 9900 \times 100 = 85.83\%$ .

**Final:**  $E_b \approx 209.1 \text{ V}$ ,  $T_a \approx 91.4 \text{ N}\cdot\text{m}$ , output  $\approx 8.50 \text{ kW}$ , efficiency  $\approx 85.8\%$ .

## Part C — Group 5

### C5(A). Three-point and four-point starters

A starter inserts resistance in series with armature at starting because  $E_b = 0$ . The handle moves over studs to cut out resistance as speed rises.

**Three-point:** Terminals L, A, F. OLR is in armature path and trips on excess current. NVR is in series with the shunt field and holds the handle at RUN. Supply failure or field opening releases it. Field weakening may reduce NVR current and cause unwanted release.

**Four-point:** Terminals L, A, F, N. NVR is connected across supply independently, so field weakening does not release the handle. It needs separate field-failure protection. Diagrams must show resistance studs, handle, spring, NVR and OLR.



### C5(B). Load test and Swinburne's test

**Direct load test:** Couple motor to brake/dynamometer, start safely, increase load, record  $V$ ,  $I$ ,  $N$  and torque. Calculate input  $VIL$ , output  $2\pi NT/60$  and efficiency. It gives actual temperature, commutation and efficiency but wastes full-load power and needs loading equipment.

**Swinburne's test:** Run shunt motor no-load at rated voltage. Measure  $V$ ,  $I_0$ ,  $I_{sh}$  and  $R_a$ . Determine constant rotational loss from no-load input after subtracting field and no-load armature copper loss. Predict efficiency at any load by adding load copper losses.

**Advantages:** economical, simple, low power, predicts motor/generator efficiency. **Limitations:** no full-load temperature/commutation test, inaccurate stray-load effects, not suitable for series motors.

## Part C — Group 6

### C6(A). Speed-control methods

From  $N \propto (V - I_a R_a) / \Phi$ :

**Shunt field control:** increase field resistance  $\rightarrow$  reduce  $I_f$  and  $\Phi \rightarrow$  speed above base. Efficient but torque capacity falls.

**Shunt armature resistance:** add series resistance  $\rightarrow$  larger  $I_a R$  drop  $\rightarrow$  lower speed. Simple but inefficient and load-dependent.

**Voltage control:** vary armature voltage with constant field  $\rightarrow$  smooth efficient speed control below base.

**Series motor:** field diverter weakens field and raises speed; armature diverter strengthens effective field and lowers speed; tapped-field control changes turns; series resistance gives low-speed starting control.

Diagrams should show the rheostat/diverter position and current paths. Flux control is generally above base; armature voltage/resistance is below base.

### C6(B). Traction control and braking

At starting, two series motors are connected in series with maximum starting resistance. As speed and back EMF rise, resistance is cut out to full-series. The motors are then transitioned to parallel, first with limiting resistance and finally full-parallel.

**Shunt transition:** temporary shunt resistors maintain current while reconnection occurs. **Bridge transition:** a bridge network gives smoother current/torque continuity but uses more complex contactors.

**Rheostatic braking:** motors generate into resistors; kinetic energy becomes heat. **Regenerative braking:** generated EMF exceeds line voltage with correct polarity, returning current and energy to the supply. Mechanical brakes remain necessary for low speed, parking and emergency.

# Practice answers and final numerical values

## A. Fill in the blanks

1 Yoke. 2 Two. 3 Critical field resistance. 4 Zero. 5  $\Phi I_a$ . 6 No-volt. 7 Braking resistor/rheostat. 8 Full.

## B. True or false

1 False—lap winding is preferred for low-voltage high-current duty. 2 True. 3 False—series motor no-load speed may become dangerous. 4 True. 5 True. 6 True.

## C. MCQs

1-C, 2-B, 3-B, 4-B, 5-C, 6-B, 7-C, 8-B.

## D. Part-identification and connection answers

1. Pole shoe.
2. Commutator.
3. In a shunt generator, the field is across terminal voltage;  $I_a$  reaches the brush junction and divides into  $I_L$  and  $I_{sh}$ , so  $I_a = I_L + I_{sh}$ .
4. The three-point starter has L to starting resistance/armature through OLR, F through the NVR/field circuit and a spring-return handle.
5. Series: supply  $\rightarrow$  resistance  $\rightarrow$  M1  $\rightarrow$  M2  $\rightarrow$  return. Parallel: both M1 and M2 form separate branches across the same supply.

## E. Characteristic-curve answers

1. External voltage includes both armature-reaction reduction and internal resistive/brush drops; falling shunt voltage also weakens field current.
2. The knee marks onset of magnetic saturation; beyond it, large field-current increase gives small voltage increase.
3. At light current, series-field flux is small. Since  $N \propto E_b / \Phi$ , speed rises sharply.
4. Over-compounded cumulative generator, because its rising terminal voltage can compensate feeder drop.

## F. Numerical answers

| No. | Working summary                                              | Final answer     |
|-----|--------------------------------------------------------------|------------------|
| 1   | $E_g = 4 \times .02 \times 800 \times 600 / (60 \times 2)$   | 320 V            |
| 2   | $\Phi = 60 \times 6 \times 300 / (6 \times 900 \times 1000)$ | 0.02 Wb = 20 mWb |
| 3   | $I_a = 50 + 2 = 52$ A; $E_g = 250 + 52 \times .2$            | 260.4 V          |
| 4   | $I_a = (220 - 205) / .3$                                     | 50 A             |
| 5   | $T = 9550 \times 10 / 750$                                   | 127.3 N·m        |
| 6   | $N_2 = 900 / 0.75$                                           | 1200 rpm         |
| 7   | $R_{st} = 240 / 60 - .4$                                     | 3.6 $\Omega$     |
| 8   | $I = 360 / 3$ ; $P = 360 \times 120$                         | 120 A, 43.2 kW   |

## G. Motor-selection answers

1. **Lathe:** shunt or separately excited motor because speed regulation is good and speed can be controlled smoothly.
2. **Crane:** series motor because very high starting torque and overload capacity are required.
3. **Conveyor:** cumulative compound motor because it combines high starting torque with better speed regulation than a series motor.
4. **Efficient below-base control:** armature-voltage control with constant field.
5. **Energy recovery:** regenerative braking, provided the supply can accept returned power.

### REFERENCES AND ONLINE RESOURCES

## Official syllabus references

### Textbooks and reference books

| Code | Book                                                                                                                                           |
|------|------------------------------------------------------------------------------------------------------------------------------------------------|
| T1   | B. L. Theraja and A. K. Theraja, <i>Electrical Technology</i> , Volume II, S. Chand & Co.                                                      |
| T2   | B. L. Theraja and A. K. Theraja, <i>Electrical Technology</i> , Volume III, S. Chand & Co.                                                     |
| R1   | P. S. Bimbhra, <i>Electrical Machinery</i> , Khanna Publishers.                                                                                |
| R2   | N. V. Suryanarayana, <i>Utilization of Electric Power: Including Electric Drives and Electric Traction</i> , New Age International Publishers. |
| R3   | K. Murugesh Kumar, <i>DC Machines and Transformers</i> , S. Chand & Company.                                                                   |
| R4   | J. B. Gupta, <i>Theory and Performance of Electrical Machines</i> , S. K. Kataria & Sons.                                                      |
| R5   | M. V. Deshpande, <i>Electrical Machines</i> , Prentice Hall India, New Delhi, 2011.                                                            |
| R6   | I. J. Nagrath and D. P. Kothari, <i>Electrical Machines</i> , Tata McGraw-Hill Education.                                                      |
| R7   | B. R. Gupta, <i>Fundamentals of Electric Machines</i> , Third Edition, New Age International Publishers.                                       |

### Official online resources listed in the syllabus

|                                      |                   |                                              |                   |
|--------------------------------------|-------------------|----------------------------------------------|-------------------|
| NPTEL 108102146                      | <a href="#">↗</a> | NPTEL 108105155                              | <a href="#">↗</a> |
| NPTEL 108106071                      | <a href="#">↗</a> | NPTEL 108105053                              | <a href="#">↗</a> |
| SWAYAM                               | <a href="#">↗</a> | IIT Bombay Virtual Labs                      | <a href="#">↗</a> |
| COEP Electrical Machines Virtual Lab | <a href="#">↗</a> | IIT Guwahati Virtual Electrical Machines Lab | <a href="#">↗</a> |
| NPTEL YouTube Channel                | <a href="#">↗</a> | Electrical Machines Tutorial                 | <a href="#">↗</a> |



### Resource note

External websites require internet access. The core handbook, formulas, diagrams, animation controls and print/PDF function work offline.

## FINAL EXAMINATION CHECKLIST

# Before submitting the answer book

### Numerical answers

- ✓ Write given data with units.
- ✓ Convert mWb to Wb and kW to W where needed.
- ✓ Identify lap/wave and correct A.
- ✓ Use Ia rather than IL where appropriate.
- ✓ State brush-drop and loss assumptions.
- ✓ Box the final answer with unit.

### Theory and diagrams

- ✓ Start with definition/principle.
- ✓ Draw labeled diagram with arrows and current path.
- ✓ Explain construction and working in sequence.
- ✓ Label graph axes and operating regions.
- ✓ Give advantages, limitations and applications where asked.
- ✓ End with the required conclusion or comparison.

### Seven-mark answer structure

Definition/principle → labeled diagram → construction/connection → step-by-step working → formula/curve → advantages/limitations → application or conclusion.